African Livestock and Crop Health Guide

Signs and symptoms, prevention, and treatment of diseases, pests and parasites - with labelled diagrams

LIVESTOCK: Chickens | Goats | Pigs | Cattle

CROPS: Tomatoes | Cabbage | Maize | Beans | Sorghum

Inside this guide

- Species and crop health basics, vital signs and biosecurity
- Nine labelled body and plant diagrams showing where to look
- 47 livestock and 48 crop problems: cause, signs, prevention, treatment
- Notifiable and zoonotic disease warnings, and when to call a vet
- Safe pesticide and animal-drug use, plus an outbreak action plan

IMPORTANT

This guide is for education and first-line farm guidance. It does not replace a veterinarian, animal health worker or agricultural extension officer. Always confirm a diagnosis, follow product labels and national laws, and report notifiable diseases (e.g. ASF, FMD, PPR, Newcastle, avian influenza, anthrax) to the veterinary authorities.

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1. How to Use This Guide

Read this first - it makes every later page easier to use.

This guide helps farmers, animal health workers, students and extension staff in African countries to **recognise** common diseases, pests and parasites of four livestock species and five food crops, to **prevent** them, and to take sensible first steps in **treatment and control**. Each problem is set out in the same way: **cause and spread, signs and symptoms, prevention, and treatment and control**.

Labelled diagrams. Every species and crop has a full-page diagram with numbered parts. Look at the diagram for the animal or plant, go through each numbered part, then read the list beneath it: it tells you what an abnormal sign looks like and which disease it may point to. Then read the matching disease entry for details.

Golden rules for farm health

- Prevention is cheaper than cure: vaccinate on schedule, control ticks, flies and worms, keep housing clean and dry, feed well, and provide clean water.
- Isolate any sick animal or plant immediately and quarantine all new stock (animals 2-3 weeks, plants 1-2 weeks) before mixing.
- Several different diseases look alike. Use this guide to narrow the possibilities, and ask a vet, animal health worker or extension officer to confirm - especially before spending money on drugs or chemicals.
- Report suspected notifiable diseases (marked NOTIFIABLE) to your District Veterinary or Agricultural Officer at once.
- Use veterinary drugs and pesticides ONLY as the product label and a vet/extension officer direct: correct dose, correct species, withdrawal period (animals) and pre-harvest interval (crops).
- Wear gloves, boots and a mask when handling sick animals, carcasses, pesticides or aborted material, and wash hands afterwards.

Zoonotic diseases - those that can pass from animals to people

Some diseases in this guide can infect people: **anthrax, brucellosis, Rift Valley fever, avian influenza, orf, bovine tuberculosis, swine erysipelas, pork tapeworm (cysticercosis)** and others. Never eat or sell meat or milk from sick animals or animals found dead; boil or pasteurise milk; cook meat thoroughly; and never open a carcass suspected of anthrax.

Normal vital signs of healthy adult animals

Animal	Temperature	Heart rate / min	Breaths / min	Notes
Chicken	40.6-43.0 °C	-	-	Bright eyes, red comb, alert, eats and drinks
Goat	38.5-40.0 °C	70-90	12-30	Rumen moves 1-1.5 times a minute
Pig	38.0-39.5 °C	60-90	10-30	Piglets run slightly higher
Cattle	38.0-39.5 °C	48-84	10-30	Chewing the cud most of the day

Figures are approximate adult values and change with age, season, pregnancy and stress. A temperature 1 °C or more above normal usually indicates infection. Use a rectal thermometer for mammals.

Biosecurity and general health management

Biosecurity checklist for any livestock farm

- Fence or house animals; control visitors, vehicles, dogs, wildlife and rodents. Provide a footbath and handwashing point at the entrance.
- Quarantine new animals for at least 2-3 weeks and examine them before they join the herd or flock.
- Buy animals from known healthy sources, not from the nearest market where possible; avoid sharing bulls, boars, bucks or equipment.
- Do not feed swill, kitchen waste or raw meat scraps to pigs; do not allow free-range pigs to scavenge near latrines.
- Clean and disinfect pens, houses, drinkers and feeders regularly; remove manure; keep litter and floors dry.
- Dispose of dead animals promptly by deep burial with lime or burning; do not dump carcasses in water or bush; do not skin or eat animals found dead.
- Keep records: vaccinations, treatments, births, deaths and movements - these are useful for vets and for disease control.
- Vaccinate on schedule against the diseases present in your area, and keep vaccines cold (2-8 °C) in a cool box or refrigerator.

Responsible use of veterinary drugs and acaricides

- Use the correct dose for the animal's bodyweight and species; under-dosing builds resistance in worms, ticks and bacteria.
- Complete the course; do not share or reuse needles; observe the withdrawal period before selling milk, eggs or meat.
- Rotate classes of dewormers and acaricides; buy drugs only from registered veterinary outlets; check expiry dates and keep out of reach of children.
- Antibiotics do not work on viral diseases (e.g. ND, PPR, ASF, FMD, LSD); they only treat secondary bacterial infection.

Safe use of pesticides and fungicides on crops

- Identify the problem first; use only products registered in your country for that crop and pest, in the label dose.
- Wear gloves, long clothing, boots, mask and eye protection; never mix or spray near water sources, food or children.
- Observe the pre-harvest interval and re-entry period; store chemicals locked away in original containers; triple-rinse and puncture empty containers.
- Use integrated pest management (IPM): resistant varieties, crop rotation, clean seed, field hygiene, traps, natural enemies and botanicals such as neem, with chemicals only when needed.

Quick reference: notifiable and zoonotic diseases

Disease	Species	Why it matters
African swine fever	Pigs	No vaccine or treatment; kills up to 100 percent; report and cull
Foot-and-mouth disease	Cattle, goats, pigs	Extremely contagious; stops trade; vaccinate with matching serotypes
Peste des petits ruminants	Goats, sheep	High deaths; one-dose vaccine gives long protection
Newcastle disease / avian influenza	Chickens	Flock wipe-outs; avian influenza can infect people
Anthrax	Cattle, goats	Sudden death; do not open carcass; deadly to people
Rift Valley fever	Cattle, goats, people	Abortion storms after rains; people infected by blood and milk
Lumpy skin disease, CBPP	Cattle	Notifiable; vaccinate and control movement
Brucellosis, bovine TB	Cattle, goats	Spread to people through raw milk; test and cull
Taenia solium (pork measles)	Pigs, people	Causes epilepsy in people; use latrines and cook pork well

What to do in a suspected disease outbreak (livestock)

- 1. Isolate sick animals at once and stop all movement of animals, products and people in and out of the farm.
- 2. Call your vet or District Veterinary Officer; give number sick and dead, species, age, signs and recent movements.
- 3. Do not slaughter, sell or eat sick animals or animals that die suddenly; do not move carcasses.
- 4. Wear gloves and boots; wash and disinfect hands, boots, equipment and vehicles after contact.
- 5. Dispose of carcasses by deep burial with lime or burning, as directed by officials.
- 6. Follow official orders on quarantine, vaccination and compensation; restock only after cleaning and the recommended rest period.

What to do in a suspected crop disease or pest outbreak

- 1. Walk the field in a zig-zag and record how many plants are affected and which parts (leaf, stem, root, fruit, grain).
- 2. Take a sample (whole plant with roots, wrapped in damp paper) to the extension officer or plant clinic; photograph symptoms.
- 3. Remove and destroy badly infected plants; do not compost diseased material.
- 4. Choose control according to the cause: fungicide for fungi, bactericide or hygiene for bacteria, vector control for viruses, insecticide or biocontrol for insects.
- 5. Clean tools, boots and wheelbarrows between fields; plan crop rotation and resistant varieties for the next season.

PART A - LIVESTOCK DISEASES, PESTS AND PARASITES

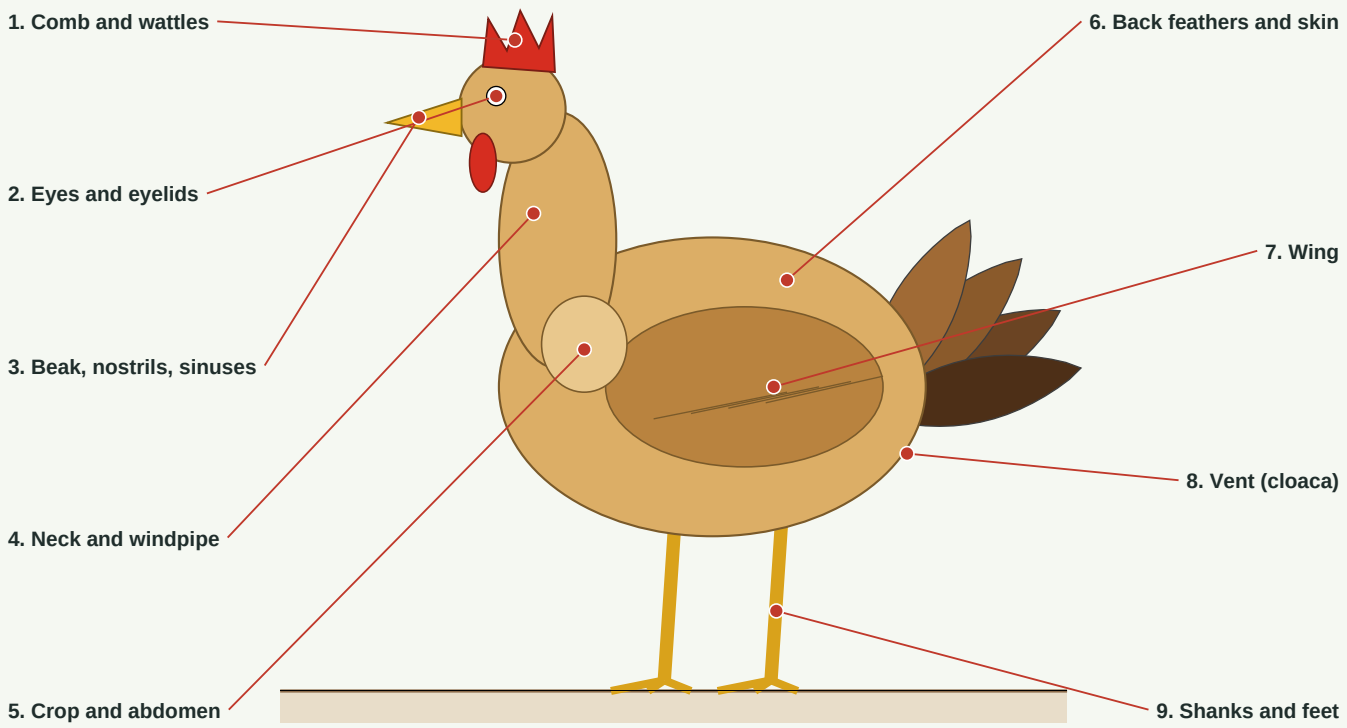
Chickens, goats, pigs and cattle

Livestock health in Africa is shaped by heat, rain, tick and tsetse pressure, free-range management, shared grazing and markets. The sections below follow the same order: a labelled diagram, then each disease or parasite. Start with the diagram to decide **where** the problem is, then read the entries.

A1. Chickens (Poultry)

Most village flocks are free-range and unvaccinated. Newcastle disease, parasites and predators are the leading causes of loss.

Figure 1. Chicken - where to look for disease



A healthy chicken has a bright red comb, clear eyes, clean nostrils and vent, smooth feathers and straight, scaly-free shanks. Numbers match the table below.

No.	Body part / plant part	What to look for (and likely causes)
1	**Comb and wattles**	Pale or blue-purple (fowl cholera, anaemia from mites); white crusty or scabby warts (fowl pox)
2	**Eyes and eyelids**	Swollen, sticky, closed eyelids; foamy eyes; grey or cloudy iris (Marek's); pox scabs (fowl pox)
3	**Beak, nostrils, sinuses**	Nasal discharge, sneezing, gaping, facial swelling (Newcastle, coryza, gapeworm, pox mouth lesions)
4	**Neck and windpipe**	Twisted neck, tremors, rattling or wheezing breath (Newcastle disease, bronchitis)
5	**Crop and abdomen**	Empty or sour crop; swollen belly (ascites); worms and weight loss
6	**Back feathers and skin**	Lice, mites, scabs, feather loss, scratching; heat-stress or skin pox lesions
7	**Wing**	Drooping or paralysed wing (Marek's, Newcastle); bleeding or pox lesions
8	**Vent (cloaca)**	Pasted, bloody, white or green droppings (coccidiosis, pullorum, Newcastle); mites and lice
9	**Shanks and feet**	Scaly-leg mite crusts, swollen joints, leg paralysis; bleeding spots (avian influenza)

1. Newcastle Disease (ND)

Viral - NOTIFIABLE - most common cause of village flock wipe-outs

CAUSE AND SPREAD	Virulent avian paramyxovirus. Spreads by bird-to-bird contact, droplets, contaminated feed, water, cages, vehicles, shoes, live-bird markets and apparently healthy carrier birds. Found in all African countries.
SIGNS AND SYMPTOMS	• Sudden deaths, sometimes with no earlier signs • Greenish, watery diarrhoea; ruffled feathers; drooping wings; sleepy, huddled birds • Gasping, coughing, sneezing, rattling breath; swollen head and eyelids • Nervous signs: twisted neck (torticollis), tremors, circling, walking backwards, paralysis of legs or wings • Sudden drop in egg laying; thin-shelled, rough or misshapen eggs • Mortality can reach 80-100 percent in unvaccinated flocks
PREVENTION	• Vaccinate: live LaSota or HB1 vaccine in drinking water or eye-drop; thermostable I-2 vaccine by eye-drop is suited to village flocks. Repeat every 3-4 months as advised by the vet or extension officer • Quarantine new birds for 2 weeks; do not mix species, ages or birds from markets • Restrict visitors; footbath and handwashing at the coop door • Clean and disinfect houses, feeders and drinkers between flocks
TREATMENT AND CONTROL	• No cure; antibiotics only control secondary bacterial infection • Isolate sick birds; give electrolytes and vitamins in clean water as supportive care • Report suspected outbreaks to the District Veterinary Officer • Bury dead birds deeply with lime or burn them; do not sell or eat sick birds • Vaccinate healthy neighbouring birds only on veterinary advice

2. Infectious Bursal Disease (Gumboro)

Viral - affects chicks 3 to 6 weeks old

CAUSE AND SPREAD	Birnavirus that destroys the bursa of Fabricius (the immune organ), so birds become easily infected by other diseases. Very hardy virus that survives in houses for months; spreads by contaminated litter, equipment and people.
SIGNS AND SYMPTOMS	• Sudden outbreak: many chicks sick within 1-2 days • Ruffled feathers, huddling, dullness, trembling, reluctance to move • Whitish or watery diarrhoea; soiled vent; birds peck at their own vent • Dehydration; deaths peak on days 3-4 then fall • Survivors are weak, stunted and have poor vaccine response (immune suppression)
PREVENTION	• Vaccinate with an intermediate Gumboro vaccine at 10-14 days, booster at 21-28 days (follow hatchery advice) • Buy chicks from breeder flocks vaccinated so chicks have maternal antibodies • All-in, all-out stocking; wash, disinfect and rest houses 2 weeks between batches • Keep different ages separate
TREATMENT AND CONTROL	• No specific treatment • Supportive: clean water with electrolytes and glucose, vitamins A, E and K • Keep chicks warm and reduce stress • Antibiotics only for secondary infections, as advised by a vet

3. Fowl Pox

Viral - spread by mosquitoes

CAUSE AND SPREAD	Avipoxvirus entering through skin wounds, mosquito and mite bites or contact. Common in the rainy season and where there is stagnant water.
SIGNS AND SYMPTOMS	• Dry form: wart-like, yellow then dark scabs on comb, wattles, eyelids, beak corners and unfeathered legs • Wet (diphtheritic) form: yellow, cheesy patches in mouth, tongue, throat and windpipe • Difficulty eating, swallowing and breathing; birds may gape • Eyelids swell shut; reduced growth and egg production
PREVENTION	• Vaccinate by wing-web stab from 6-8 weeks, before the mosquito season; check for a 'take' (small scab) after 7-10 days • Drain stagnant water; use mosquito control; keep houses clean and dry • Isolate sick birds; do not mix vaccinated and unvaccinated flocks
TREATMENT AND CONTROL	• No cure; disease runs 3-4 weeks and recovered birds are immune • Gently remove scabs and touch lesions with iodine or antiseptic • Vitamin A and C in water; soft food for birds with mouth lesions • Antibiotics for secondary bacterial infections as advised by a vet

4. Marek's Disease

Viral - tumour disease of growing birds 6 to 20 weeks

CAUSE AND SPREAD	Herpesvirus shed in feather dander and dust, carried on wind and equipment. Birds are infected as chicks but show signs later.
SIGNS AND SYMPTOMS	• Paralysis of one or both legs: one leg stretched forward, one backward ('doing the splits') • Drooping wings, twisted neck, difficulty standing • Grey or cloudy eye (grey iris) and irregular pupil; blindness • Enlarged, hard feather follicles and skin lumps; tumours in organs • Weight loss, pale comb, deaths
PREVENTION	• Vaccinate day-old chicks (HVT vaccine) at the hatchery - this is the only effective control • Keep brooding chicks away from older birds; strict hygiene • Buy from reputable hatcheries
TREATMENT AND CONTROL	• No treatment; infected birds remain carriers • Cull paralysed or sick birds humanely and dispose of them safely

5. Coccidiosis

Protozoal - very common in damp litter

CAUSE AND SPREAD	Eimeria parasites in droppings. Birds swallow oocysts from contaminated litter, water and feed. Most severe in chicks 3-8 weeks old, in wet and overcrowded conditions.
SIGNS AND SYMPTOMS	• Bloody or brown droppings (caecal form) or watery pale droppings • Ruffled feathers, huddling, closed eyes, pale comb and wattles • Poor appetite and thirst; weight loss; poor growth • Pasted, dirty vents; sudden deaths in severe outbreaks
PREVENTION	• Keep litter dry; fix leaking drinkers; avoid overcrowding • Use coccidiostat in starter feed or vaccinate (ask the feed supplier or vet) • Clean drinkers and feeders often; rest and disinfect houses; rotate free-range yards • Control rodents and wild birds
TREATMENT AND CONTROL	• Amprolium or sulphonamide (e.g. sulfadimethoxine) or toltrazuril in drinking water - follow the product label and treat for 3-5 days • Vitamin K and vitamin A to reduce bleeding and help recovery • Replace wet litter, provide clean water and electrolytes • Observe the withdrawal period before selling eggs or meat

6. Fowl Cholera

Bacterial (Pasteurella multocida)

CAUSE AND SPREAD	Spread by sick birds, wild birds, rodents, contaminated water and equipment. Occurs in acute and chronic forms.
SIGNS AND SYMPTOMS	• Acute: sudden deaths in apparently healthy birds; high fever; greenish-yellow diarrhoea • Swollen, bluish-purple comb and wattles; laboured breathing • Chronic: swollen wattles filled with cheesy pus, swollen joints, lameness, nasal discharge
PREVENTION	• Strict biosecurity; keep out wild birds and rodents; use clean water • All-in, all-out production; do not add new birds to an infected flock • Vaccinate where the disease is endemic (vaccines available from vets)
TREATMENT AND CONTROL	• Sulphonamides or tetracyclines in water, or injectable antibiotics, as prescribed by a vet • Cull chronic carriers; clean and disinfect the house

7. Pullorum Disease and Fowl Typhoid

Bacterial (Salmonella Pullorum / Gallinarum) - passed through eggs

CAUSE AND SPREAD	Infected breeder hens pass Salmonella to chicks inside the egg. Also spreads through contaminated litter, feed and carrier birds.
SIGNS AND SYMPTOMS	• Pullorum (chicks 0-3 weeks): white chalky diarrhoea pasting the vent, huddling near heat, gasping, many deaths in the first 2 weeks • Fowl typhoid (older birds): pale comb, yellow-green diarrhoea, fever, thirst, loss of appetite, drop in egg production, deaths • Dead-in-shell embryos and weak chicks
PREVENTION	• Buy chicks from Salmonella-tested hatcheries; blood-test and cull reactor breeders • Hygiene and disinfection; control rodents and flies • Vaccinate against fowl typhoid where recommended (9R vaccine)
TREATMENT AND CONTROL	• Antibiotics (sulphonamides, quinolones, etc.) reduce deaths only on a vet's prescription, but survivors stay carriers • Cull the whole infected breeder flock where possible

8. Highly Pathogenic Avian Influenza (HPAI)

Viral - NOTIFIABLE - ZOONOTIC (can infect people)

CAUSE AND SPREAD	Influenza A virus (H5 and H7 types) introduced by wild waterfowl, live-bird trade and contaminated equipment; spreads very fast between birds.
SIGNS AND SYMPTOMS	• Sudden high deaths in many birds within 24-48 hours • Swollen head and face; purple or blue comb and wattles • Bleeding spots on the shanks (legs) and under the skin • Respiratory distress, diarrhoea, nervous signs, sharp drop in eggs
PREVENTION	• Keep poultry away from wild birds and water bodies; cover feed and drinkers • Do not buy birds of unknown origin; avoid sharing equipment or crates • Wear gloves and a mask when handling sick or dead birds; wash hands thoroughly
TREATMENT AND CONTROL	• DO NOT TREAT - report immediately to the veterinary authorities • Authorities will cull (stamping out), disinfect and control movement • Do not slaughter, sell or eat sick birds; do not move birds out of the area

9. External Parasites: Mites, Lice, Fleas and Scaly-leg Mite

Parasitic - common in free-range and deep-litter houses

CAUSE AND SPREAD	Red mites hide in cracks, perches and nest boxes and bite at night; northern fowl mites and lice live on the bird; sticktight fleas cling to the head; scaly-leg mites burrow under shank scales.
SIGNS AND SYMPTOMS	• Restless birds at night, pale comb (anaemia), reluctance to sit in nest boxes • Scratching, feather loss, scabs, dirty or broken feathers; lice and eggs at the feather base and vent • Red or grey specks of mites on perches and eggs • Scaly leg: raised, crusty, thickened scales on the shanks; lameness • Drop in egg production and weight loss
PREVENTION	• Provide a dust-bathing area with sand and wood ash • Seal cracks, clean and replace litter and nesting material regularly; use metal or plastic perches • Inspect birds and perches weekly; quarantine new birds • Do not overcrowd
TREATMENT AND CONTROL	• Spray or dust birds and house with an approved product (e.g. permethrin or cypermethrin spray, carbaryl dust) following the label; treat cracks and perches; repeat after 7-10 days • Scaly leg: soak legs in warm soapy water then apply petroleum jelly, sulphur ointment or approved product • Observe label withdrawal times for eggs and meat

10. Internal Parasites: Roundworms, Tapeworms and Gapeworm

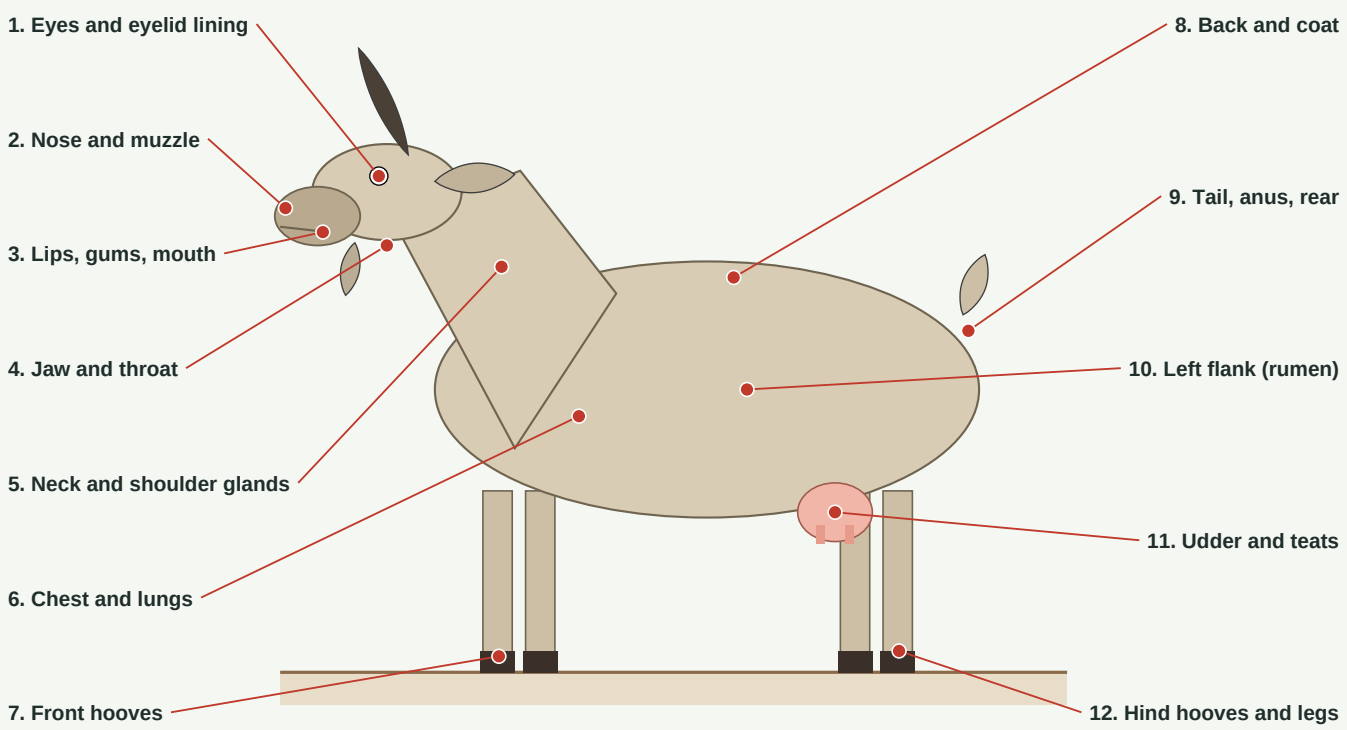
Parasitic - worm burden is highest in free-range birds

CAUSE AND SPREAD	Birds swallow worm eggs from contaminated soil, litter, snails, beetles or earthworms. Gapeworm (Syngamus) lives in the windpipe.
SIGNS AND SYMPTOMS	• Weight loss, poor growth, pale comb and wattles • Diarrhoea, soiled vents, sometimes worms visible in droppings • Fall in egg production • Gapeworm: gaping, head shaking, coughing and gasping (looks like a respiratory disease)
PREVENTION	• Keep litter dry; rotate runs; separate young and old birds • Deworm regularly (every 2-3 months for free-range flocks) and before laying • Control intermediate hosts (snails, earthworms, beetles); clean feeders and drinkers
TREATMENT AND CONTROL	• Piperazine, levamisole, fenbendazole or albendazole in feed or water - follow the label • Repeat after 2-3 weeks; give vitamins to help recovery • Observe withdrawal periods for eggs and meat

A2. Goats

Goats are hardy but suffer badly from PPR, worms, pneumonia, ticks and foot problems.

Figure 2. Goat - where to look for disease



Check the eyelid colour (FAMACHA), jaw, nose, mouth, flanks, udder, feet and rear end of every goat regularly.

No.	Body part / plant part	What to look for (and likely causes)
1	**Eyes and eyelid lining**	Pale or white inner eyelid = anaemia (FAMACHA 4-5, haemonchosis); sticky discharge (PPR)
2	**Nose and muzzle**	Crusty or purulent nasal discharge (PPR, CCPP, pneumonia); scabs (orf)
3	**Lips, gums, mouth**	Ulcers and erosions (PPR), scabby sores (orf), blisters (FMD), foul breath
4	**Jaw and throat**	'Bottle jaw' soft swelling under the jaw (stomach worms, liver fluke)
5	**Neck and shoulder glands**	Swollen lymph nodes; lumps or abscesses (caseous lymphadenitis)
6	**Chest and lungs**	Cough, fast or painful breathing, stands with neck extended (CCPP, pneumonia)
7	**Front hooves**	Lameness, foul-smelling interdigital rot, overgrown hooves (foot rot, foot scald)
8	**Back and coat**	Scabs, hair loss, ticks, lice (mange); rough coat (worms)
9	**Tail, anus, rear**	Soiled, pasted tail and watery diarrhoea (PPR, coccidiosis, worms)
10	**Left flank (rumen)**	Swollen left flank = bloat or enterotoxaemia; empty flank = not eating
11	**Udder and teats**	Hot, hard, swollen quarter or abnormal milk (mastitis); scabs on teats (orf)
12	**Hind hooves and legs**	Lameness, swollen joints (CAE), foot rot

1. Peste des Petits Ruminants (PPR) - 'Goat Plague'

Viral - NOTIFIABLE - very high mortality

CAUSE AND SPREAD	Morbillivirus of goats and sheep. Spreads by close contact, coughing and discharges, and through markets, shared water and trade animals. Found across Africa.
SIGNS AND SYMPTOMS	• High fever (40-41 °C), dullness, loss of appetite • Watery then thick, sticky yellow discharge from eyes and nose, crusting and sealing the eyelids and nostrils • Ulcers and erosions on gums, tongue and inside the lips; foul breath, drooling • Severe, foul-smelling, watery diarrhoea and dehydration • Coughing and rapid breathing (pneumonia); abortion; death within 5-10 days, especially in kids
PREVENTION	• Vaccinate - a single dose of live attenuated PPR vaccine gives long (often lifelong) immunity; vaccinate kids from about 3-4 months • Quarantine new animals for 3 weeks; avoid mixing flocks at markets and water points • Report outbreaks and restrict movements
TREATMENT AND CONTROL	• No antiviral cure • Supportive care: antibiotics (e.g. oxytetracycline) for secondary pneumonia, oral rehydration, clean eyes, nose and mouth gently, good feed and shelter • Isolate sick animals and report to veterinary services

2. Contagious Caprine Pleuropneumonia (CCPP)

Bacterial (Mycoplasma capricolum capripneumoniae)

CAUSE AND SPREAD	Spread by coughing and close contact; a highly contagious lung disease of goats found in drier regions across Africa.
SIGNS AND SYMPTOMS	• High fever, severe cough, rapid and painful breathing • Standing with neck stretched, elbows out; reluctance to move • Nasal discharge, frothy saliva; weight loss • High deaths in adults and kids
PREVENTION	• Vaccinate (CCPP vaccine, annual) in endemic areas • Quarantine new goats; avoid contact with infected herds and shared grazing
TREATMENT AND CONTROL	• Early treatment: tylosin, long-acting oxytetracycline, florfenicol or tilmicosin as prescribed by a vet • Isolate sick goats; provide shelter and clean water • Report suspected outbreaks

3. Haemonchosis (Barber's Pole Worm) and Other Stomach Worms

Parasitic - the biggest production killer of goats in Africa

CAUSE AND SPREAD	Worm larvae are eaten with grass. Haemonchus contortus sucks blood in the abomasum; worm eggs are passed in dung and hatch rapidly in warm, wet weather.
SIGNS AND SYMPTOMS	• Pale or white eyelid lining and gums (anaemia) - FAMACHA score 4-5 means treat • 'Bottle jaw': soft swelling under the lower jaw; sometimes swollen belly • Weakness, rough coat, weight loss, poor milk production, lagging behind the herd • Dark or scouring droppings; sudden death in kids and heavily infected animals
PREVENTION	• Use the FAMACHA eye chart to select animals for treatment (targeted selective treatment) • Rotate grazing; do not overstock; avoid grazing very wet or short pasture; allow browsing • Do not underdose - this builds resistance • Quarantine and deworm new animals; use faecal egg counts to monitor
TREATMENT AND CONTROL	• Use an effective dewormer (albendazole, fenbendazole, levamisole, closantel or ivermectin) at the correct dose for goat bodyweight; goats usually need a higher dose than sheep or cattle of some products - check with the vet or label • Give good protein feed, iron and vitamin B12 to anaemic animals • Observe the milk and meat withdrawal period

4. Heartwater

Rickettsial (Ehrlichia ruminantium) - tick-borne (Amblyomma ticks)

CAUSE AND SPREAD	Carried by bont ticks. Affects goats, sheep and cattle in many parts of Africa; new animals brought into tick areas are most at risk.
SIGNS AND SYMPTOMS	• Sudden high fever and loss of appetite • Nervous signs: walking in circles, high-stepping gait, twitching eyelids, chewing movements, convulsions • Fluid in the chest and around the heart; laboured breathing • Sudden death, sometimes with no earlier signs
PREVENTION	• Control ticks by dipping or spraying with an approved acaricide • Allow young animals to be infected early in tick-endemic areas where this is advised by the vet • Be careful when introducing animals from tick-free areas
TREATMENT AND CONTROL	• Early treatment with oxytetracycline (high dose, long-acting) as prescribed by the vet • Supportive care; keep animals calm in shade with clean water

5. Orf (Contagious Ecthyma)

Viral - ZOONOTIC (people can catch it)

CAUSE AND SPREAD	Parapoxvirus spreading by contact and scabs left on fences, feeding troughs and thorny browse. Common in kids and lambs.
SIGNS AND SYMPTOMS	• Blisters then thick, crusty scabs on lips, nostrils, muzzle and around the mouth • Lesions on teats of does and sometimes on feet or genitals • Kids refuse to suckle and lose weight; sore mouth
PREVENTION	• Vaccinate with live scab vaccine where available; avoid abrasive browse • Quarantine new animals; clean, disinfect and rest contaminated pens • Wear gloves when handling sick animals to avoid infection
TREATMENT AND CONTROL	• Self-limiting in 2-4 weeks; no specific treatment • Clean lesions gently with antiseptic; apply protective ointment; give soft feed • Antibiotics for secondary infection; fly repellent on open sores • Hand-feed kids that cannot suckle

6. Foot Rot and Foot Scald

Bacterial (Dichelobacter nodosus, Fusobacterium)

CAUSE AND SPREAD	Occurs in wet, muddy conditions and overgrown hooves; spreads from infected hooves to others on shared soil.
SIGNS AND SYMPTOMS	• Lameness, kneeling to graze, reluctance to walk • Red, swollen, moist skin between toes; foul smell • Hoof horn separates and peels away; pus under the sole
PREVENTION	• Trim hooves regularly; keep floors dry and well-drained • Footbath (10 percent zinc sulphate) once a week during the wet season • Quarantine and inspect feet of newly bought goats; vaccinate where available
TREATMENT AND CONTROL	• Trim away damaged horn; clean wound • Apply topical oxytetracycline spray or antiseptic; soak feet in zinc sulphate or copper sulphate footbath • Injectable long-acting oxytetracycline or penicillin in severe cases (vet) • Keep animals on clean, dry ground

7. Enterotoxaemia (Overeating Disease)

Bacterial (Clostridium perfringens types C and D)

CAUSE AND SPREAD	Gut bacteria multiply and release toxin after a sudden change to rich feed, lush pasture or excess grain.
SIGNS AND SYMPTOMS	• Sudden death of the best-fed, fastest-growing kids • Abdominal pain, bloating, kicking at belly • Profuse diarrhoea; convulsions and paddling
PREVENTION	• Vaccinate with multivalent clostridial vaccine (kids from 4-6 weeks, booster 4 weeks later; adults annually and before kidding) • Change feed gradually over 7-10 days; avoid overfeeding grain; feed fibre
TREATMENT AND CONTROL	• Antitoxin and penicillin if caught early; the course is often too rapid • Reduce feed, provide fluids, call a vet

8. Coccidiosis in Kids

Protozoal (Eimeria)

CAUSE AND SPREAD	Kids pick up oocysts from dirty, wet, overcrowded pens and contaminated feeding areas. Most common at 3-8 weeks and at weaning.
SIGNS AND SYMPTOMS	• Diarrhoea (may be dark or bloody), straining, soiled tail • Dehydration, rough coat, poor growth, weight loss • Death in severe cases
PREVENTION	• Keep pens clean and dry; raise feeders and water troughs off the floor • Avoid overcrowding and stress at weaning; provide coccidiostat in creep feed if advised
TREATMENT AND CONTROL	• Amprolium, sulphonamide or toltrazuril as per label or vet • Oral rehydration solution; separate and keep warm

9. Mange, Lice and Ticks

Parasitic - skin

CAUSE AND SPREAD	Sarcoptes or Psoroptes mites burrow in skin; lice live in the coat; ticks feed on blood and spread disease. Spread by contact and shared housing.
SIGNS AND SYMPTOMS	• Intense itching and rubbing; scabs, crusts and hair loss especially on face, ears, legs, belly • Thickened, wrinkled skin; weight loss and poor condition • Lice: restlessness, dull coat. Ticks: visible ticks on ears, udder, groin; anaemia
PREVENTION	• Regular dipping, spraying or pour-on of the herd • Quarantine and treat new goats; clean and disinfect housing; avoid overcrowding
TREATMENT AND CONTROL	• Ivermectin injection (repeat after 10-14 days) or amitraz or pyrethroid sprays and washes per label • Treat the whole herd and the pens; separate heavily infested animals • Observe withdrawal periods for milk and meat

10. Mastitis

Bacterial (Staphylococcus, Streptococcus, E. coli)

CAUSE AND SPREAD	Bacteria enter through the teat canal from dirty bedding, unclean hands or injury.
SIGNS AND SYMPTOMS	• Hot, swollen, hard, painful half of the udder • Watery, clotted, bloody or yellow milk; reduced milk yield • Fever and loss of appetite; kids refuse to suckle • In severe cases the udder turns blue-black and cold (gangrenous mastitis)
PREVENTION	• Wash hands and udder; dry-wipe teats before milking; dip teats after milking • Keep bedding clean and dry; milk sick animals last; dry-doe treatment • Cull chronic cases
TREATMENT AND CONTROL	• Strip out the affected side often; apply warm compresses • Intramammary antibiotic infusion and injectable antibiotics as prescribed by the vet; anti-inflammatory • Discard milk during treatment and withdrawal period

11. Brucellosis

Bacterial (Brucella melitensis) - ZOONOTIC

CAUSE AND SPREAD	Spread through aborted foetuses, placenta, birth fluids and milk. Humans get 'Malta fever' from raw milk or contact.
SIGNS AND SYMPTOMS	• Abortion in the last third of pregnancy; retained placenta • Weak kids; reduced milk; swollen testicles in bucks; joint swelling and lameness
PREVENTION	• Test and cull positive animals; vaccinate kids (Rev-1) where the national programme allows • Boil or pasteurise milk; wear gloves and boots when handling aborted material • Isolate aborting does; dispose of foetuses and placenta safely
TREATMENT AND CONTROL	• No practical treatment; cull infected animals under veterinary guidance

12. Bloat (Ruminal Tympany)

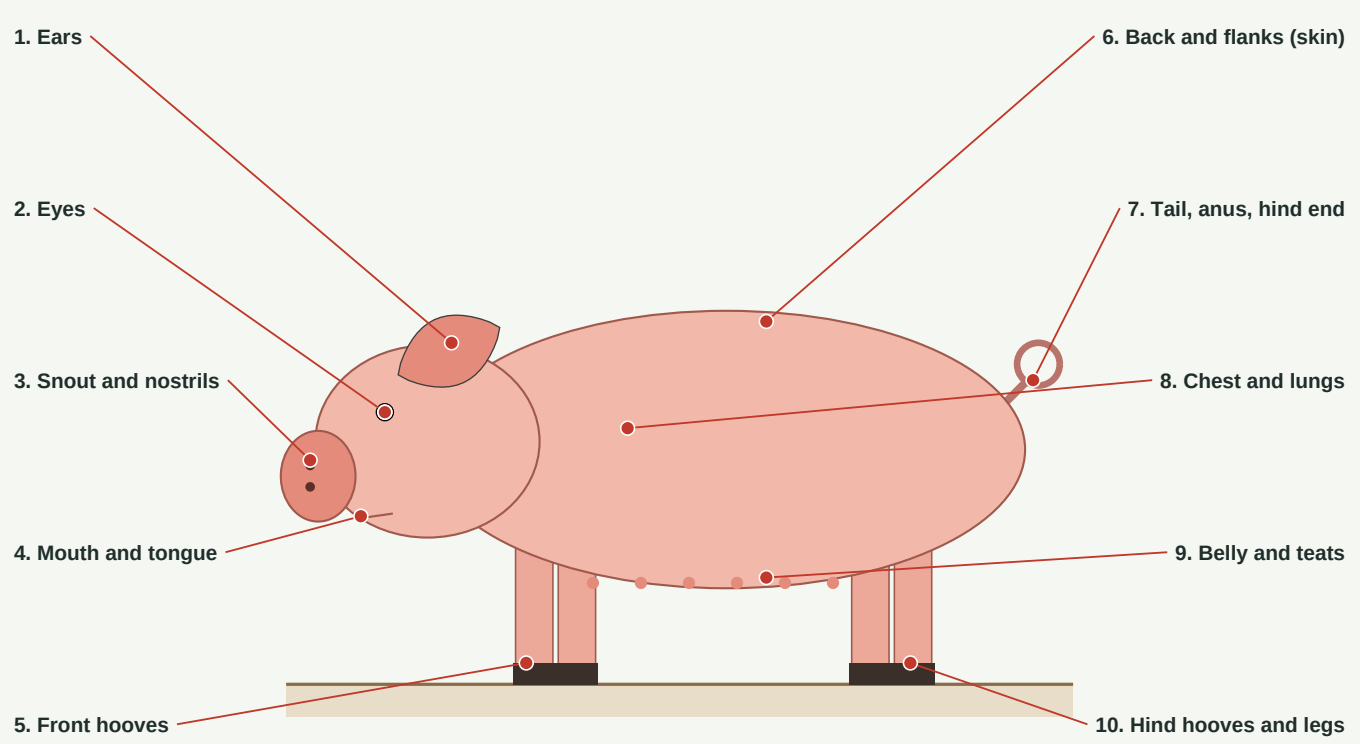
Metabolic - emergency

CAUSE AND SPREAD	Gas builds up in the rumen after eating lush wet legumes, grain, or after choking.
SIGNS AND SYMPTOMS	• Swollen left flank, distress, grunting, kicking at the belly • Difficulty breathing, salivation; collapse and death within hours
PREVENTION	• Do not turn hungry goats onto lush legumes or wet pasture; feed dry hay first • Introduce grain and new feeds gradually
TREATMENT AND CONTROL	• Stand the animal with front legs raised; walk it • Give vegetable oil or an anti-bloat drench; pass a stomach tube to release gas • In emergencies a vet may puncture the rumen

A3. Pigs

Pig farming is growing fast; African swine fever and poor biosecurity are the biggest threats.

Figure 3. Pig - where to look for disease



Examine ears, skin, snout, eyes, belly and feet. Many pig diseases show first on the ears and belly skin.

No.	Body part / plant part	What to look for (and likely causes)
1	**Ears**	Red or purple blotches (ASF, classical swine fever); crusting and scabs; ticks
2	**Eyes**	Reddish eyes with sticky discharge (swine fever); swollen eyelids (oedema disease)
3	**Snout and nostrils**	Discharge, sneezing, twisted snout; blisters (FMD)
4	**Mouth and tongue**	Blisters or ulcers (FMD); drooling; cysts under tongue (pork measles)
5	**Front hooves**	Lameness, blisters on the coronary band (FMD), overgrown or cracked hooves
6	**Back and flanks (skin)**	Diamond-shaped red lesions (erysipelas); scabs and itching (mange); hair loss
7	**Tail, anus, hind end**	Diarrhoea, blood in dung, rectal prolapse; sore, bitten tail
8	**Chest and lungs**	Cough, laboured abdominal breathing, fever (pneumonia, ASF)
9	**Belly and teats**	Purple discolouration of belly (ASF); mastitis in sows; wet or soiled underline
10	**Hind hooves and legs**	Swollen joints, lameness (arthritis), foot lesions

1. African Swine Fever (ASF)

Viral - NOTIFIABLE - no vaccine or treatment (as at time of writing)

CAUSE AND SPREAD	DNA virus that spreads by contact with sick pigs, warthogs and bushpigs, soft ticks (Ornithodoros), contaminated pork, swill feeding, vehicles, boots and equipment. Survives long in pork and the environment.
SIGNS AND SYMPTOMS	• High fever (40-42 °C); loss of appetite; huddling; weakness • Red to purple blotches on ears, snout, belly, inner legs and tail • Vomiting, bloody diarrhoea, laboured breathing, cough • Abortion in sows; sudden deaths - mortality up to 100 percent in unvaccinated herds • Cannot be reliably distinguished from classical swine fever without laboratory tests
PREVENTION	• Strict biosecurity: pen pigs; no free-range scavenging; fence off warthogs and bushpigs • Never feed swill, kitchen waste or pork scraps unless boiled for 30 minutes • Limit visitors; footbath and clean clothes; buy pigs only from known healthy sources and quarantine for 3 weeks • Disinfect vehicles, crates and pens; control soft ticks in pens and rodents
TREATMENT AND CONTROL	• There is no treatment or widely approved vaccine - report immediately to the veterinary authorities • Stamping out: authorities cull infected herds, burn or deep-bury carcasses, disinfect, and enforce movement control and a rest period before restocking • Do not sell, slaughter, butcher or move sick pigs or pork; never dump carcasses in water or bush

2. Classical Swine Fever (Hog Cholera)

Viral - NOTIFIABLE

CAUSE AND SPREAD	Pestivirus spread by contact, contaminated feed and equipment, and carrier pigs; looks similar to ASF.
SIGNS AND SYMPTOMS	• High fever, huddling, loss of appetite • Red, inflamed eyes with sticky discharge; constipation then yellow diarrhoea • Purple blotches on skin; staggering gait; tremors in piglets • Abortions, stillbirths; chronic wasting
PREVENTION	• Vaccinate with a licensed live vaccine where permitted; biosecurity as for ASF • Quarantine new pigs; no swill feeding
TREATMENT AND CONTROL	• No treatment - notify the veterinary authorities; cull and dispose of infected pigs

3. Foot-and-Mouth Disease in Pigs

Viral - NOTIFIABLE

CAUSE AND SPREAD	Highly contagious; pigs spread huge amounts of virus. See also the cattle section.
SIGNS AND SYMPTOMS	• Fever; blisters on the snout, tongue, lips, coronary band and between the toes, and on the teats • Severe lameness; pigs refuse to stand; hooves may be lost • Piglets can die suddenly
PREVENTION	• Vaccinate with a vaccine matching local serotypes; restrict movement and visitors • Do not feed untreated waste from infected areas
TREATMENT AND CONTROL	• No specific treatment; supportive care and soft bedding • Report to authorities; isolate and follow official control instructions

4. Swine Erysipelas

Bacterial (Erysipelothrix rhusiopathiae) - ZOONOTIC

CAUSE AND SPREAD	Bacteria live in soil and on tonsils of healthy pigs; outbreaks follow stress, weather changes or poor hygiene.
SIGNS AND SYMPTOMS	• Sudden fever (41 °C), stiffness, reluctance to move • Raised, red or purple diamond-shaped skin patches on the back, sides and neck • Swollen painful joints (chronic arthritis); abortion; sudden death
PREVENTION	• Vaccinate breeding stock and growers as per vet schedule • Keep pens clean; control rodents; reduce stress
TREATMENT AND CONTROL	• Penicillin early, as prescribed by the vet; anti-inflammatory for fever • People: wear gloves; the infection can cause skin lesions (erysipeloid) in humans

5. Internal Parasites: Ascaris, Nodular Worm and Others

Parasitic

CAUSE AND SPREAD	Eggs passed in dung survive for months in pens and soil; pigs swallow them from contaminated floors, feed and water. Free-range and wet pens have the highest worm burden.
SIGNS AND SYMPTOMS	• Poor growth and feed efficiency, pot belly, rough coat • Coughing ('thumps') when larvae migrate through the lungs • Diarrhoea; worms visible in dung; 'milk spot' liver damage at slaughter
PREVENTION	• Wash sows before farrowing; clean and dry pens; do not overstock • Deworm sows 1-2 weeks before farrowing, piglets at weaning, and growers every 2-3 months • Rotate pastures for outdoor pigs
TREATMENT AND CONTROL	• Ivermectin, fenbendazole, albendazole or levamisole as per label; repeat as directed • Observe withdrawal period before slaughter

6. Mange and Lice

Parasitic - skin

CAUSE AND SPREAD	Sarcoptes scabiei mites burrow in skin; spread from sow to piglets and between pigs in contact; lice live on skin folds.
SIGNS AND SYMPTOMS	• Intense itching; rubbing on walls and posts • Scabs and thickened, crusty skin on ears, flanks, legs and belly; hair loss • Reduced growth and appetite; sows scratching
PREVENTION	• Treat sows before farrowing; treat new pigs on arrival • Clean and disinfect pens; avoid overcrowding
TREATMENT AND CONTROL	• Ivermectin injection (repeat in 10-14 days) or amitraz wash/spray per label • Treat the whole herd and the pens • Observe withdrawal periods

7. Porcine Cysticercosis (Pork Measles)

Parasitic (Taenia solium) - ZOONOTIC - public health danger

CAUSE AND SPREAD	Pigs eat human faeces containing tapeworm eggs when free-ranging near homes. Cysts develop in pig muscle and tongue. People eating undercooked infected pork get tapeworms; eggs shed by carriers can cause neurocysticercosis (epilepsy) in humans.
SIGNS AND SYMPTOMS	• Pigs usually look healthy • Small white cysts (measles) in the tongue, meat and heart; cysts may be felt under the tongue • Discovered at slaughter or meat inspection
PREVENTION	• Build and use latrines; keep pigs penned, not free-roaming • Cook pork thoroughly; condemn infected carcasses at meat inspection • Treat people with tapeworms; community health education
TREATMENT AND CONTROL	• Oxfendazole may be used in pigs under vet advice, but carcasses of infected pigs must not be eaten • Human carriers are treated by health workers (e.g. niclosamide or praziquantel)

8. Neonatal and Post-weaning Diarrhoea (E. coli Scours)

Bacterial

CAUSE AND SPREAD	Piglets pick up E. coli from dirty farrowing pens, udders and floors; cold, damp pens and poor colostrum intake increase risk.
SIGNS AND SYMPTOMS	• Watery yellow, white or grey diarrhoea; wet, dirty rear end; strong odour • Dehydration, sunken eyes, shivering; piglets lose weight and die • Post-weaning: scours, poor growth, swollen eyelids (oedema disease)
PREVENTION	• Clean and disinfect farrowing pens; wash sows before farrowing • Ensure piglets suckle colostrum within hours of birth • Keep the piglet nest warm (about 30-35 °C), dry and draught-free • Vaccinate sows where available; avoid sudden weaning feed changes
TREATMENT AND CONTROL	• Oral rehydration solution and warmth • Antibiotics chosen by the vet (preferably after sensitivity testing) • Keep sick piglets separate and pens clean

9. Piglet Anaemia (Iron Deficiency)

Nutritional

CAUSE AND SPREAD	Sow milk is low in iron; piglets on concrete floors cannot get iron from soil.
SIGNS AND SYMPTOMS	• Pale skin, wrinkled hairless skin; poor growth at 1-3 weeks • Rapid, laboured breathing ('thumping'); weak piglets; sudden death
PREVENTION	• Inject iron dextran as per label within the first 3 days of birth; repeat if advised • Provide clean iron-rich supplements or a sod of clean soil in concrete pens
TREATMENT AND CONTROL	• Iron injection and good nutrition; treat the sow and piglets as per vet advice

10. Swine Pneumonia (Mycoplasma and Bacterial)

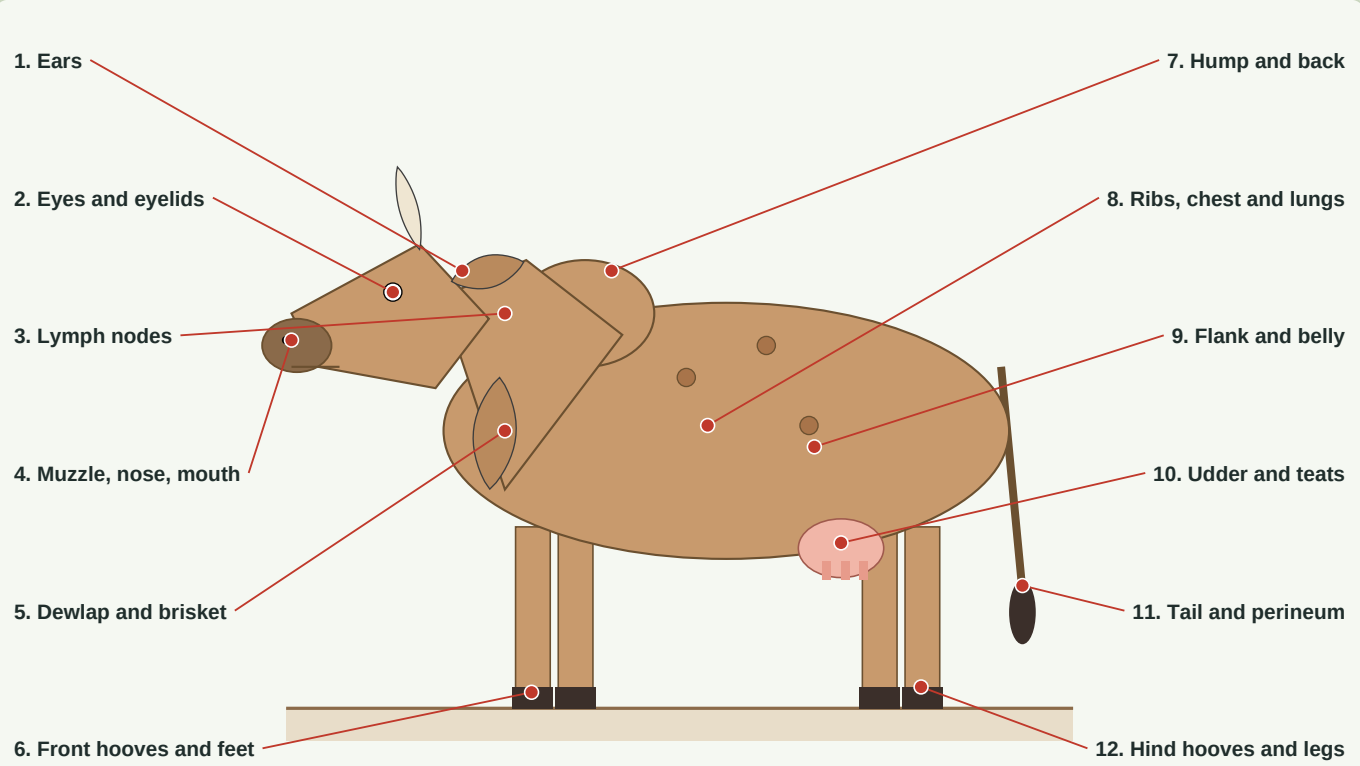
Bacterial

CAUSE AND SPREAD	Spread by coughing and nose-to-nose contact; worsened by cold, damp, poor ventilation, overcrowding and dust.
SIGNS AND SYMPTOMS	• Dry or moist cough; fever; laboured, abdominal breathing • Poor growth, dullness, reduced appetite; sometimes sudden deaths
PREVENTION	• Good ventilation, avoid draughts and overcrowding; all-in, all-out • Vaccinate against Mycoplasma and bacteria where advised; quarantine new pigs
TREATMENT AND CONTROL	• Antibiotics prescribed by a vet (e.g. tylosin, tetracyclines, florfenicol) • Isolate sick pigs; give warmth and fresh air

A4. Cattle

Tick-borne diseases, trypanosomiasis, FMD, LSD and worms cause the largest cattle losses in Africa.

Figure 4. Cattle - where to look for disease



Regular inspection of ears, lymph nodes, eyes, skin, udder and feet allows early detection and treatment.

No.	Body part / plant part	What to look for (and likely causes)
1	**Ears**	Brown ear ticks clustered inside the ear (ECF risk); wounds and ear-tag injury
2	**Eyes and eyelids**	Watery or pale eyes (ECF, anaemia); pale or yellow lining (redwater, anaplasmosis)
3	**Lymph nodes**	Hard swollen lymph nodes in front of the shoulder and below the ear (ECF, TB, LSD)
4	**Muzzle, nose, mouth**	Drooling, blisters and erosions on gums, tongue and lips (FMD); nasal discharge
5	**Dewlap and brisket**	Swelling (bottle jaw, trypanosomiasis, heart disease); ticks
6	**Front hooves and feet**	Blisters, foul smell, lameness (FMD, foot rot, interdigital dermatitis)
7	**Hump and back**	Firm skin nodules (LSD); ticks, wounds, mange; blackleg swelling at shoulder
8	**Ribs, chest and lungs**	Cough, laboured breathing, elbows turned out (CBPP, pneumonia, TB)
9	**Flank and belly**	Swollen left flank (bloat); sunken flank (not eating); pot belly (worms)
10	**Udder and teats**	Hot hard quarter, clots in milk (mastitis); blisters (FMD); ticks on udder
11	**Tail and perineum**	Soiled with diarrhoea or blood; ticks under the tail; aborted foetus discharge
12	**Hind hooves and legs**	Lameness, swollen joints, foot lesions; blackleg swelling in thigh

1. Foot-and-Mouth Disease (FMD)

Viral - NOTIFIABLE - also affects goats, sheep and pigs

CAUSE AND SPREAD	Aphthovirus with several serotypes (O, A, SAT 1, SAT 2 and SAT 3 in Africa). Spreads by breath, contact, milk, shared water and grazing, and carrier buffalo; extremely contagious.
SIGNS AND SYMPTOMS	• Fever, loss of appetite, dullness • Blisters (vesicles) on the tongue, gums, dental pad, lips, muzzle and teats; blisters then burst leaving raw ulcers • Heavy drooling, ropy saliva, smacking lips • Blisters between the hooves and on the coronary band; lameness, reluctance to walk • Sharp drop in milk yield; calves may die suddenly from heart damage
PREVENTION	• Vaccinate every 6 months in endemic areas with a vaccine matching local serotypes • Movement control; quarantine new animals; avoid shared grazing with buffalo • Disinfect vehicles, boots and equipment; clean water points
TREATMENT AND CONTROL	• No specific treatment; report immediately to veterinary services • Supportive care: soft feed, clean water, wash mouth with mild antiseptic, clean and dress foot lesions, fly repellent • Antibiotics for secondary infections as prescribed by a vet • Isolate sick animals; follow official control measures

2. East Coast Fever (ECF)

Protozoal (Theileria parva) - tick-borne (brown ear tick)

CAUSE AND SPREAD	Transmitted by Rhipicephalus appendiculatus, the brown ear tick. A major killer of cattle in eastern, central and southern Africa (Kenya, Uganda, Tanzania, Malawi, Zambia, Zimbabwe and neighbours).
SIGNS AND SYMPTOMS	• High fever (40-42 °C), loss of appetite, dullness • Greatly swollen lymph nodes in front of the shoulder (prescapular) and below the ear (parotid) • Watery eyes, nasal discharge, laboured breathing, frothy discharge from nose just before death • Diarrhoea and weight loss; death within 2-3 weeks
PREVENTION	• Tick control: dipping or spraying every 5-7 days in high tick pressure; rotate acaricide types to prevent resistance • Infection-and-treatment (ITM) vaccination programme where available • Do not introduce animals without tick control; keep pastures rested
TREATMENT AND CONTROL	• Early treatment with buparvaquone (or parvaquone) by a vet or trained animal health worker - treat early for the best results • Supportive care; keep affected animals in shade with water • Continue tick control

3. Babesiosis (Redwater) and Anaplasmosis (Gall Sickness)

Blood parasite and bacterial - tick-borne

CAUSE AND SPREAD	Babesia bigemina/bovis and Anaplasma marginale are carried by blue ticks (Rhipicephalus decoloratus and R. microplus). Anaplasma can also be spread by biting flies and contaminated needles.
SIGNS AND SYMPTOMS	• High fever, weakness, pale or yellow mucous membranes (anaemia, jaundice) • Redwater: red, coffee-coloured urine; Anaplasmosis: jaundice but no red urine • Constipation, loss of appetite, rapid breathing, abortions • Cerebral babesiosis: aggression, staggering, convulsions
PREVENTION	• Tick control (dipping, spraying); use clean needles on every animal • Vaccinate where live blood vaccines are available • Maintain endemic stability by letting calves be infected while protected by maternal immunity
TREATMENT AND CONTROL	• Diminazene aceturate or imidocarb for babesiosis; long-acting oxytetracycline or imidocarb for anaplasmosis - as prescribed by a vet • Blood transfusion and supportive treatment in severe anaemia • Rest the animal; avoid forced exercise

4. Heartwater in Cattle

Rickettsial - tick-borne (bont tick)

CAUSE AND SPREAD	Same disease as in goats (see Goats section). Often seen in animals moved into tick areas.
SIGNS AND SYMPTOMS	• Sudden high fever, loss of appetite • Nervous signs: circling, high stepping, convulsions, champing jaws • Laboured breathing, diarrhoea; sudden death
PREVENTION	• Control bont ticks with acaricides; manage introduction of naive animals • Infection-and-treatment method where advised by the vet
TREATMENT AND CONTROL	• Early treatment with long-acting oxytetracycline as prescribed • Supportive care

5. Lumpy Skin Disease (LSD)

Viral (capripoxvirus) - NOTIFIABLE

CAUSE AND SPREAD	Spread mainly by biting insects (stable flies, mosquitoes) and ticks, also by contaminated equipment and animal movement. Widespread in Africa; outbreaks follow rains.
SIGNS AND SYMPTOMS	• Fever, enlarged superficial lymph nodes, reduced appetite and milk • Firm, round, raised skin nodules 2-5 cm across, on head, neck, body, udder and genitals • Nodules may ulcerate, scab and become infected; lesions in mouth, nose and eyes • Swelling of legs and brisket; lameness; abortion and temporary or permanent infertility
PREVENTION	• Annual vaccination with a live attenuated capripox (Neethling-type) vaccine - vaccinate all in-contact cattle • Control biting flies and ticks; avoid moving cattle during outbreaks; quarantine new animals • Report suspected cases and disinfect equipment
TREATMENT AND CONTROL	• No specific antiviral treatment • Supportive: antibiotics for secondary infections, anti-inflammatory drugs, wound cleaning, fly repellent, soft feed • Isolate sick animals; follow veterinary authority guidance

6. Contagious Bovine Pleuropneumonia (CBPP)

Bacterial (Mycoplasma mycoides) - NOTIFIABLE

CAUSE AND SPREAD	Spread by coughing between close contact animals; chronic carriers keep the disease in herds. Present in many sub-Saharan countries.
SIGNS AND SYMPTOMS	• Fever, cough, nasal discharge • Rapid, shallow breathing; stands with elbows out and neck stretched • Painful chest; weight loss; swollen joints in calves • Deaths, mostly in adult cattle
PREVENTION	• Vaccination (T1/44 or T1SR) as part of national programmes • Test and slaughter; movement control; quarantine new animals
TREATMENT AND CONTROL	• Antibiotic treatment is discouraged because treated animals can become carriers - follow veterinary authority instructions • Report to the veterinary authorities

7. Anthrax

Bacterial (Bacillus anthracis) - NOTIFIABLE - ZOONOTIC - emergency

CAUSE AND SPREAD	Spores live in the soil for decades; cattle swallow them while grazing, especially in dry, dusty seasons or after floods. People can be infected by handling or eating meat of dead animals.
SIGNS AND SYMPTOMS	• Sudden death, often with no previous signs • Bloody discharge (dark, unclotted blood) from nose, mouth, anus; bloated carcass; no rigor mortis • Fever, staggering, trembling, breathing difficulty and collapse before death in less sudden cases
PREVENTION	• Annual vaccination (Sterne spore vaccine) before the dry or high-risk season • DO NOT open the carcass; keep dogs and scavengers away; do not drag carcasses • Burn or deep-bury the carcass with lime under supervision of the vet
TREATMENT AND CONTROL	• Early treatment with penicillin or long-acting oxytetracycline (prescribed by vet) can save animals only in early stages • Isolate and vaccinate the herd after the vet confirms the outbreak • Never skin, butcher or eat meat from animals that die suddenly; use gloves and report

8. Blackleg

Bacterial (Clostridium chauvoei)

CAUSE AND SPREAD	Spores in soil enter through the gut or wounds; occurs mostly in young, fast-growing cattle 6 months to 2 years old in good condition.
SIGNS AND SYMPTOMS	• Sudden lameness; hot, painful swelling of large muscle (hip, shoulder, thigh) • Swelling crackles when pressed (gas under the skin) and later becomes cold and painless; dark, discoloured muscle • High fever; death within 12-48 hours
PREVENTION	• Vaccinate annually with a clostridial vaccine (7-in-1 or blackleg vaccine); vaccinate calves from 3-4 months • Dispose of carcasses by burning or deep burial; avoid grazing infected pastures
TREATMENT AND CONTROL	• High-dose penicillin early, injected into and around the swelling (vet) - success is limited • Vaccinate the rest of the herd immediately

9. Brucellosis in Cattle

Bacterial (Brucella abortus) - ZOONOTIC

CAUSE AND SPREAD	Spread by aborted foetuses, placenta, vaginal discharge and milk; humans get undulant fever from raw milk and contact.
SIGNS AND SYMPTOMS	• Abortion in the last third of pregnancy; retained placenta; weak calves • Reduced milk; infertility; swollen testicles in bulls; swollen knees (hygroma)
PREVENTION	• Vaccinate heifer calves (S19 or RB51) at 3-8 months where national programmes allow • Test and slaughter positive animals; boil or pasteurise milk • Wear gloves and boots when assisting births or handling aborted material
TREATMENT AND CONTROL	• No treatment; cull positive animals under vet supervision • Isolate aborting cows; dispose of foetus and placenta safely

10. Trypanosomiasis (Nagana)

Protozoal (Trypanosoma congolense, vivax, brucei) - spread by tsetse flies

CAUSE AND SPREAD	Tsetse flies in humid, bushy areas spread the parasite; cattle in tsetse belts are most at risk. Also spread mechanically by biting flies.
SIGNS AND SYMPTOMS	• Intermittent fever; progressive anaemia (pale gums and eyelids) • Weight loss despite eating; rough staring coat; swollen lymph nodes • Swollen dewlap and brisket; weakness; reduced milk; abortion • Chronic cases die from wasting
PREVENTION	• Tsetse control: traps, targets, insecticide-treated cattle (pour-ons), clearing bush near water • Prophylaxis with isometamidium chloride injections every 3-4 months in high-risk areas • Where possible keep trypanotolerant breeds (e.g. N'Dama)
TREATMENT AND CONTROL	• Diminazene aceturate or isometamidium chloride injection according to label and vet • Supportive: good feed, iron, rest • Rotate drugs to reduce resistance; control tsetse to prevent reinfection

11. Rift Valley Fever (RVF)

Viral - NOTIFIABLE - ZOONOTIC - mosquito-borne

CAUSE AND SPREAD	Phlebovirus spread by mosquitoes after heavy rains and floods; also by contact with infected blood and aborted material. Affects cattle, sheep, goats and people.
SIGNS AND SYMPTOMS	• Abortion storms in a large number of cows at once • High deaths in newborn calves, lambs and kids • Fever, jaundice, bloody diarrhoea, nasal discharge, weakness
PREVENTION	• Vaccinate before the rainy season with a live attenuated or inactivated vaccine • Mosquito control; keep animals away from flooded areas • Do not handle aborted foetuses or sick animals without gloves; do not drink raw milk
TREATMENT AND CONTROL	• No specific treatment; supportive care • Report to authorities; isolate sick animals; dispose of aborted material safely

12. Liver Fluke (Fasciolosis)

Parasitic (Fasciola gigantica / hepatica) - snail-borne

CAUSE AND SPREAD	Flukes live in the liver; snails in swampy areas, dams and wet pastures carry the larvae, which attach to grass and are eaten by cattle.
SIGNS AND SYMPTOMS	• Chronic: weight loss, poor condition, pale mucous membranes, bottle jaw (swelling under the jaw) • Diarrhoea and reduced milk production • Acute: sudden deaths in heavily infected animals; liver condemned at slaughter
PREVENTION	• Avoid grazing swampy and wet pastures; fence off stagnant water; provide clean piped water • Control snails; strategic deworming before and after the wet season
TREATMENT AND CONTROL	• Triclabendazole (kills immature and adult flukes), or albendazole, closantel, oxyclozanide per label • Observe withdrawal periods for milk and meat

13. Ticks and Tick Control

Parasitic - skin - vector of many diseases

CAUSE AND SPREAD	Brown ear ticks, bont ticks and blue ticks feed on blood on ears, udder, groin, brisket and tail, spreading ECF, heartwater, redwater and gall sickness. Heavy burdens cause anaemia, skin damage and loss of condition.
SIGNS AND SYMPTOMS	• Visible ticks on ears, dewlap, brisket, udder, groin, perineum and under the tail • Skin wounds, scabs and secondary infection; irritation; weight loss • Anaemia, weakness, milk drop; signs of tick-borne diseases
PREVENTION	• Regular dipping, spraying or pour-on with an approved acaricide (as often as the label and local advice recommend, about every 5-7 days in heavy tick pressure) • Rotate between acaricide groups (amitraz, pyrethroids, organophosphates, macrocyclic lactones) to avoid resistance • Hand-pick ticks from calves; rotate pasture; maintain dip tanks at correct strength
TREATMENT AND CONTROL	• Wash or spray the whole animal including ears, udder and under the tail with an approved acaricide • Use gloves and protective gear when handling chemicals; follow label dose and withdrawal period • Treat tick wounds and apply wound dressing

14. Mastitis in Cattle

Bacterial (Staphylococcus, Streptococcus, E. coli)

CAUSE AND SPREAD	Bacteria enter through the teat canal, favoured by dirty housing, poor milking hygiene and teat injuries.
SIGNS AND SYMPTOMS	• Hot, swollen, hard, painful quarter; watery, clotted or bloody milk • Fever, loss of appetite, reduced milk yield • Subclinical mastitis: no visible signs but high somatic cell count / positive CMT test
PREVENTION	• Clean hands, udder and equipment; dry-wipe teats; teat dip after milking • Keep bedding and yards clean and dry; dry cow therapy; cull chronic cases • Use the California Mastitis Test (CMT) to detect early cases
TREATMENT AND CONTROL	• Strip the affected quarter frequently; intramammary antibiotic and, if needed, injectable antibiotics and anti-inflammatory as prescribed • Discard milk during treatment and withdrawal period

15. Bovine Tuberculosis (bTB)

Bacterial (Mycobacterium bovis) - ZOONOTIC

CAUSE AND SPREAD	Spread by coughing and contaminated milk; people get TB from raw milk or close contact.
SIGNS AND SYMPTOMS	• Chronic cough, weight loss and poor condition despite good feed • Enlarged lymph nodes; laboured breathing; low-grade fever
PREVENTION	• Test (tuberculin skin test) and slaughter reactors; boil or pasteurise milk • Avoid contact with infected cattle or wildlife; quarantine new cattle
TREATMENT AND CONTROL	• No practical treatment; cull positive animals under veterinary control

PART B - CROP DISEASES AND PESTS

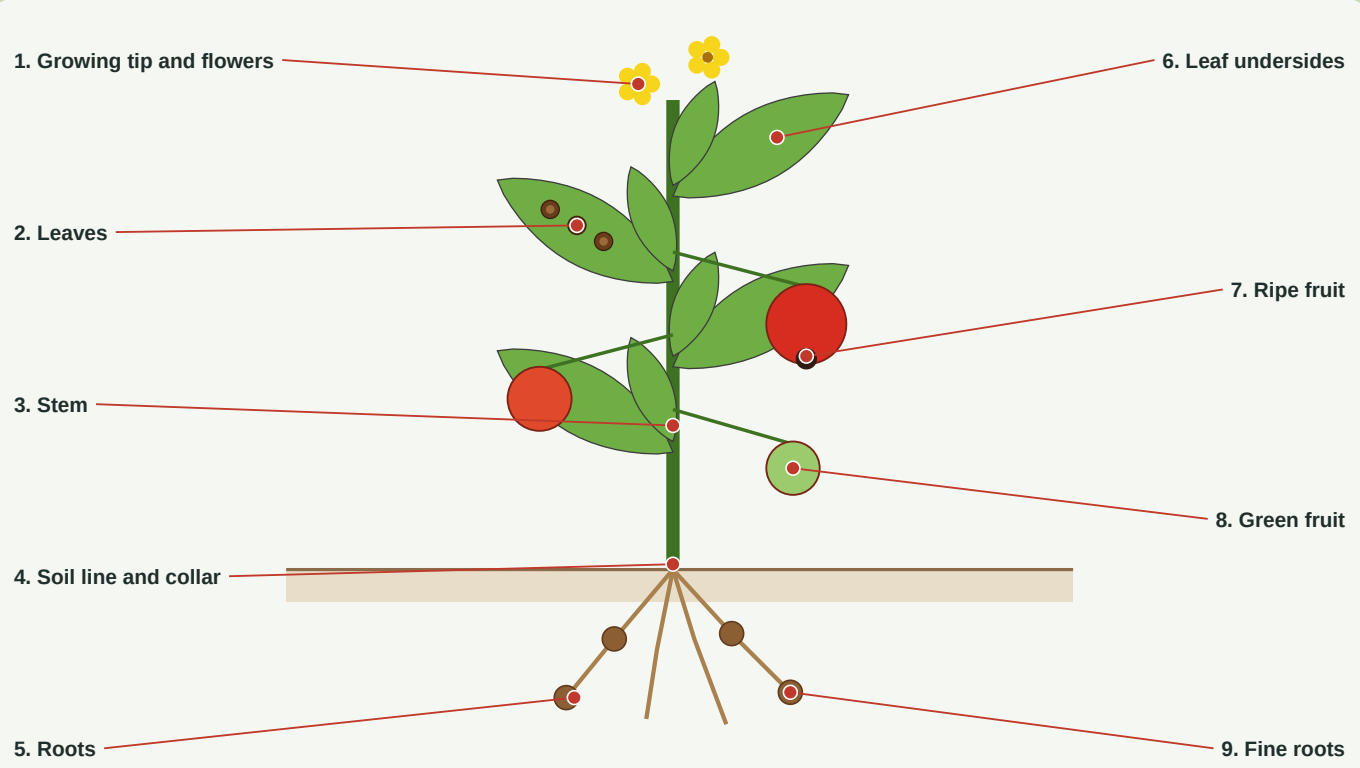
Tomatoes, cabbage, maize, beans and sorghum

Crop losses in African farming come from fungi, bacteria, viruses, insects, nematodes, parasitic weeds, birds and poor storage. Use the labelled plant diagram to find the affected part, then read the disease entries. Use clean seed, rotation, resistant varieties and good field hygiene as the foundation of control.

B1. Tomato

A high-value vegetable crop, very exposed to blight, wilt, virus and leafminer.

Figure 5. Tomato plant - where to look for disease



Inspect leaves (top and underside), stem, fruit and roots every week. Pull up one wilting plant to check the roots.

No.	Body part / plant part	What to look for (and likely causes)
1	**Growing tip and flowers**	Curled, yellow shoots (TYLCV, whitefly); flower drop; aphids and thrips
2	**Leaves**	Ringed brown spots (early blight); water-soaked patches (late blight); mines (Tuta); mosaic or curl
3	**Stem**	Dark cankers; brown inner tissue, milky ooze in water (bacterial wilt); stem borers
4	**Soil line and collar**	Collar rot, damping off; adventitious roots and wilting (bacterial wilt)
5	**Roots**	Galls or knots (root-knot nematodes); brown rotted roots (wilt)
6	**Leaf undersides**	Whiteflies, mites, aphids; mould (leaf mould); webbing
7	**Ripe fruit**	Sunken black patch at blossom end; brown greasy blotches (late blight); cracking
8	**Green fruit**	Round holes with frass (fruitworm, Tuta); spots; sunscald
9	**Fine roots**	Swollen knots (root-knot); poor root mass

1. Early Blight

Fungal (Alternaria solani)

CAUSE AND SPREAD	Fungus survives on crop debris and in soil; spreads by rain splash and wind in warm, humid conditions. Affects older leaves first.
SIGNS AND SYMPTOMS	• Dark brown spots with concentric rings ('target' or bull's-eye) on older leaves, often with a yellow halo • Leaves turn yellow, dry and drop, exposing fruit to sunscald • Dark, sunken, leathery spots with rings near the stem end of fruits • Stem lesions and collar rot on seedlings
PREVENTION	• Rotate for 2-3 years with cereals or legumes; avoid potato, pepper and eggplant • Use clean seed and healthy transplants; plant resistant or tolerant varieties • Stake and prune lower leaves; mulch; use drip irrigation; avoid wetting leaves • Remove and destroy crop debris after harvest; space plants for air flow
TREATMENT AND CONTROL	• Remove infected leaves; apply protectant fungicides such as copper-based products, mancozeb or chlorothalonil at the first signs, repeat every 7-10 days • Systemic fungicides (e.g. azoxystrobin) in rotation to prevent resistance • Follow label dose, protective gear and pre-harvest interval

2. Late Blight

Oomycete (Phytophthora infestans) - spreads very fast in cool, wet weather

CAUSE AND SPREAD	Spores travel by wind and rain from infected potato, tomato and volunteer plants; thrives at 10-25 °C with high humidity and cloud.
SIGNS AND SYMPTOMS	• Large, greasy, water-soaked green-black patches on leaves that rapidly turn brown • White fluffy mould on the underside of leaves in humid, early morning conditions • Dark brown lesions on stems; plants collapse within days • Firm, brown, greasy blotches on fruit that spread and rot
PREVENTION	• Use certified seed and healthy transplants; plant resistant varieties • Avoid overhead irrigation; space plants; remove volunteer tomato and potato plants nearby • Rotate crops; destroy infected plants and debris
TREATMENT AND CONTROL	• Apply protectant fungicides (copper, mancozeb) before wet weather; use systemic mixtures (e.g. metalaxyl-M or cymoxanil combinations) on first symptoms • Remove and burn heavily infected plants; harvest healthy fruit early • Rotate fungicide groups; follow pre-harvest intervals

3. Bacterial Wilt

Bacterial (Ralstonia solanacearum) - soil-borne, serious in warm climates

CAUSE AND SPREAD	Survives in soil and water for years; enters through roots (especially wounded by nematodes or tools). Spread by irrigation water, soil on tools, boots and infected seedlings.
SIGNS AND SYMPTOMS	• Sudden wilting of the whole plant while leaves stay green; plants may recover at night then die • Brown discolouration of stem vascular tissue when cut • Cut stem placed in a glass of clear water releases milky white bacterial streaming • Adventitious roots on the lower stem; rapid plant death
PREVENTION	• Rotate for 3-5 years with maize, sorghum or other cereals (avoid solanaceous crops, bananas and groundnuts) • Use clean seedlings from disease-free nurseries; grow on well-drained raised beds • Avoid damaging roots; disinfect tools; do not irrigate with contaminated water • Use resistant rootstock grafting or tolerant varieties; solarise soil; raise pH with lime
TREATMENT AND CONTROL	• No effective chemical cure • Uproot and burn or bury infected plants including roots and soil; do not compost • Do not move soil or water from infected plots to clean ones

4. Fusarium Wilt

Fungal (Fusarium oxysporum f. sp. lycopersici) - soil-borne

CAUSE AND SPREAD	Fungus persists in soil for many years and enters through roots; favoured by warm soil and acid, low-nutrient soils.
SIGNS AND SYMPTOMS	• Yellowing and wilting of lower leaves, often on one side of the plant or one side of a leaf • Progressive wilting up the plant; stunted growth • Brown discolouration of the vascular tissue in the stem; no bacterial ooze
PREVENTION	• Plant resistant varieties (seed labelled F or FF); use clean seed and healthy transplants • Rotate crops; solarise soil; adjust soil pH to 6.5-7 with lime; grafting on resistant rootstock • Practise good field hygiene
TREATMENT AND CONTROL	• No cure; remove and destroy diseased plants • Avoid replanting tomato or related crops in the same soil

5. Tomato Yellow Leaf Curl Virus (TYLCV)

Viral - spread by whitefly (Bemisia tabaci)

CAUSE AND SPREAD	Whiteflies feed on infected plants and spread the virus to healthy plants within minutes to hours. Seedlings infected early lose most yield.
SIGNS AND SYMPTOMS	• Leaves curl upwards, become small, yellow and thick (cup-shaped) • Stunted, bushy plants; flower drop; few small fruits • Whiteflies fly up when plants are shaken
PREVENTION	• Plant tolerant or resistant varieties • Raise seedlings under insect-proof net; use yellow sticky traps; reflective mulch • Remove weeds and infected plants; avoid planting near older infected crops; keep a crop-free period
TREATMENT AND CONTROL	• No cure for infected plants - uproot and destroy them • Control whiteflies with neem, insecticidal soap or approved insecticides (e.g. acetamiprid, imidacloprid), rotating products

6. Tomato Leafminer (Tuta absoluta)

Insect pest - caterpillars

CAUSE AND SPREAD	Tiny moths lay eggs on leaves; larvae tunnel inside leaves, stems and fruit. Multiple generations a year; very destructive.
SIGNS AND SYMPTOMS	• Blotchy, irregular mines (transparent patches) in leaves that turn brown • Tunnels in stems and growing tips; wilting shoots • Small holes in fruit calyx and fruits with frass (black droppings) • Small white, cream or pink caterpillars inside the mines
PREVENTION	• Use pheromone traps for early warning; insect nets on nurseries • Remove and destroy infested leaves, fruits and crop residue; rotate crops • Encourage natural enemies; avoid broad-spectrum sprays that kill them
TREATMENT AND CONTROL	• Spray Bacillus thuringiensis (Bt), spinosad, emamectin benzoate or chlorantraniliprole, rotating groups • Neem products; hand-remove infested leaves • Follow label pre-harvest intervals

7. Root-knot Nematodes

Nematode (Meloidogyne species) - soil pest

CAUSE AND SPREAD	Microscopic worms in warm, sandy soils feed on roots, causing galls; spread on infested soil, tools and transplants.
SIGNS AND SYMPTOMS	• Galls or swellings (knots) on roots • Stunted, yellow plants that wilt in the heat of the day; patchy fields • Poor yield and small fruits
PREVENTION	• Rotate with cereals or Tagetes (marigold) and fallow • Use resistant varieties (Mi gene) and clean transplants • Add compost; solarise soil; avoid moving soil from infested fields
TREATMENT AND CONTROL	• Remove and burn heavily galled roots after harvest; sun-dry soil • Use organic amendments such as neem cake; nematicides only where registered and under advice

8. Blossom End Rot

Physiological disorder (calcium deficiency)

CAUSE AND SPREAD	Caused by irregular watering and poor calcium movement into the fruit, not by a germ. Common in hot, dry spells followed by heavy irrigation or rain.
SIGNS AND SYMPTOMS	• Dark, sunken, leathery patch at the blossom end (bottom) of green or ripening fruit • Patch may enlarge and be invaded by moulds
PREVENTION	• Water evenly and mulch to keep soil moisture constant • Avoid excess nitrogen or ammonium fertiliser; keep soil pH 6-6.8 • Supply calcium (lime or gypsum) before planting
TREATMENT AND CONTROL	• Remove affected fruits; correct watering • Calcium nitrate foliar spray may help new fruits

9. African Bollworm / Fruitworm (Helicoverpa armigera)

Insect pest - caterpillars

CAUSE AND SPREAD	Moths lay eggs on flowers and young fruit; larvae bore into fruit and cause rot.
SIGNS AND SYMPTOMS	• Round holes bored into green fruit; caterpillars and frass inside • Fruit rots and drops; damaged flower buds and young shoots
PREVENTION	• Scout weekly; use trap crops (maize, marigold); remove infested fruits • Encourage natural enemies; avoid planting next to cotton, maize or beans at the same time
TREATMENT AND CONTROL	• Spray Bt, NPV (viral biopesticide) or neem early; spinosad or emamectin benzoate when infestation is heavy • Hand-pick caterpillars; follow pre-harvest intervals

10. Septoria Leaf Spot and Red Spider Mites

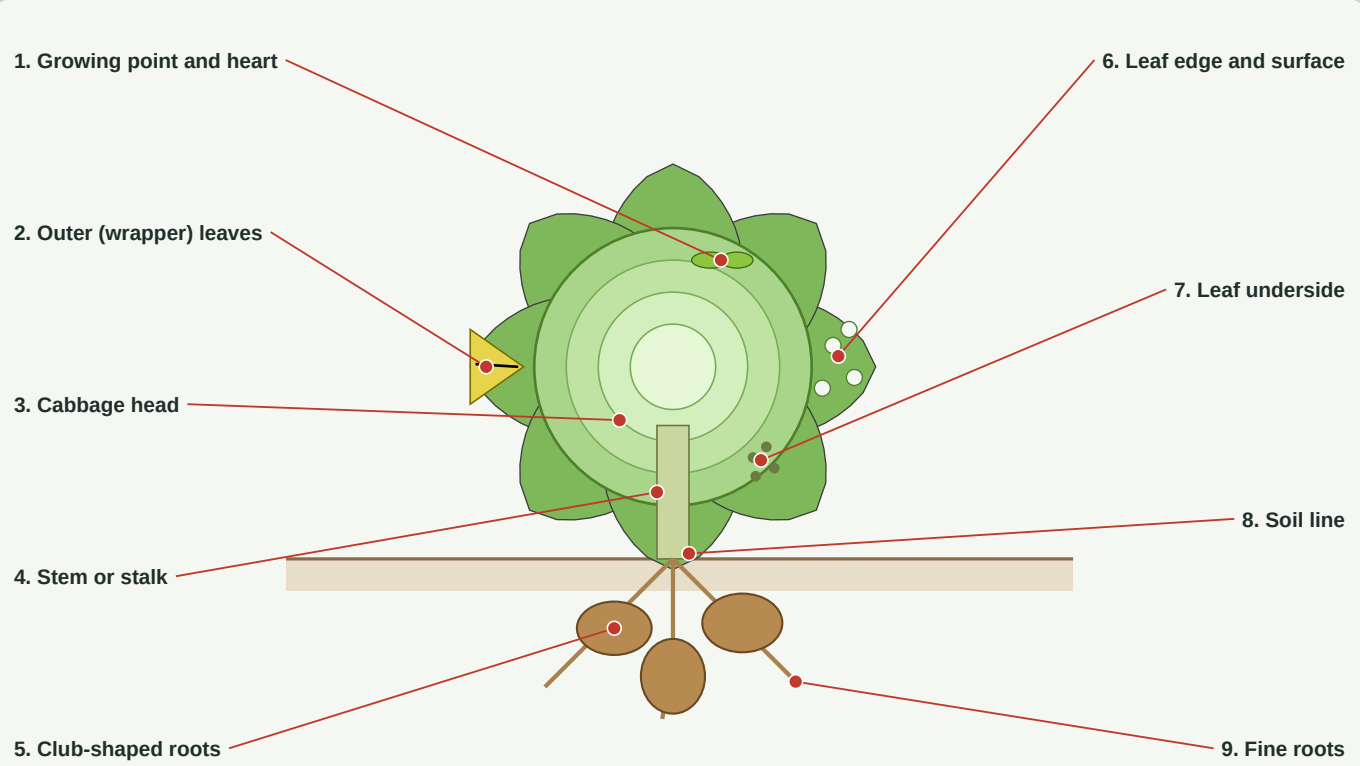
Fungal and mite pests

CAUSE AND SPREAD	Septoria spreads by rain splash from soil and debris; spider mites thrive in hot, dry, dusty conditions.
SIGNS AND SYMPTOMS	• Septoria: many small circular spots with grey-white centres and dark edges, tiny black dots in the middle; lower leaves first • Spider mites: fine webbing, pale stippling, bronzing and drying of leaves
PREVENTION	• Rotate crops; mulch; avoid wetting leaves; remove lower infected leaves • Reduce dust; avoid water stress; encourage predatory mites
TREATMENT AND CONTROL	• Septoria: copper, mancozeb or chlorothalonil sprays • Mites: wash with strong water spray, soap, neem or approved miticide; rotate products

B2. Cabbage

Brassicas are attacked by caterpillars, aphids, clubroot and bacterial diseases.

Figure 6. Cabbage - where to look for disease



Inspect outer leaves, the heart, the stem base and roots; pull up wilted plants to look for club roots.

No.	Body part / plant part	What to look for (and likely causes)
1	**Growing point and heart**	Caterpillars (diamondback moth, webworm); heart destroyed so no head forms
2	**Outer (wrapper) leaves**	V-shaped yellow lesions and blackened veins (black rot); holes (caterpillars); spots
3	**Cabbage head**	Soft, slimy, foul-smelling rot (bacterial soft rot); loose or no head; webworm tunnels
4	**Stem or stalk**	Black ring or dark vascular tissue when cut (black rot); stem rot; wirestem
5	**Club-shaped roots**	Swollen, distorted galls (clubroot); wilting at midday despite moisture
6	**Leaf edge and surface**	Windowing and shot-holes (diamondback moth); white mould (downy mildew)
7	**Leaf underside**	Grey-green aphid colonies; honeydew; sooty mould
8	**Soil line**	Seedlings cut at ground level (cutworms); damping off
9	**Fine roots**	Root maggots; poor root mass; nematode galls

1. Black Rot

Bacterial (Xanthomonas campestris pv. campestris) - seed-borne

CAUSE AND SPREAD	Bacteria enter through leaf-edge water pores and wounds; spread by splashing rain, irrigation, tools and workers; very destructive in warm, humid weather.
SIGNS AND SYMPTOMS	• V-shaped yellow lesions from the leaf edge towards the midrib • Veins turn black; leaf dries and tears • Cut stem shows black ring or dark vascular tissue • Head rots and may have a foul smell
PREVENTION	• Use clean or hot-water-treated seed (50 °C for 25 minutes) • Rotate 2-3 years with non-brassica crops; control crucifer weeds • Avoid overhead irrigation and working in wet fields; disinfect tools • Bury or remove crop residue; use resistant varieties
TREATMENT AND CONTROL	• No cure; copper-based bactericides slow spread • Remove and destroy infected plants; improve drainage

2. Clubroot

Fungus-like organism (Plasmodiophora brassicae) - acid soils

CAUSE AND SPREAD	Resting spores survive in soil for 15 years or more; spread by soil on boots, tools, water and infected seedlings. Favoured by acid, wet soils.
SIGNS AND SYMPTOMS	• Wilting at midday even when soil is moist; yellowing, stunted plants • Roots swollen into distorted, club-shaped galls • Poor or no head formation
PREVENTION	• Lime soil to pH 7.2 or higher; improve drainage • Rotate with non-brassica crops for at least 5-7 years; grow resistant varieties • Use healthy transplants and clean tools; control brassica weeds
TREATMENT AND CONTROL	• No cure - uproot and destroy plants including roots; never compost them • Do not plant brassicas in the affected field; avoid moving soil

3. Diamondback Moth (Plutella xylostella)

Insect pest - caterpillars

CAUSE AND SPREAD	Small grey moths lay eggs on leaves; green larvae wriggle when touched. Pest breeds throughout the year in warm climates and develops resistance to insecticides.
SIGNS AND SYMPTOMS	• Small holes and 'windowing' (thin transparent patches) in leaves • Larvae feeding in the heart and on the head • Silvery pupal cocoons on leaves; contaminated heads
PREVENTION	• Use pheromone traps; plant trap crops of Indian mustard; overhead sprinkler irrigation washes off larvae • Rotate with non-brassicas; remove crop residue; encourage parasitic wasps (Diadegma) • Avoid overuse of broad-spectrum insecticides
TREATMENT AND CONTROL	• Spray Bt, neem, spinosad, emamectin benzoate or indoxacarb in the evening; rotate groups • Scout weekly; spray when threshold is reached; follow pre-harvest interval

4. Cabbage Aphids (Brevicoryne brassicae)

Insect pest - sap-sucking

CAUSE AND SPREAD	Colonies of grey-green waxy aphids build up rapidly in warm, dry weather and may spread viruses.
SIGNS AND SYMPTOMS	• Clusters of grey-green aphids on leaf undersides and in the heart • Curled, yellow, distorted leaves; sticky honeydew and sooty mould • Stunted growth; head contaminated
PREVENTION	• Control weeds; avoid excess nitrogen; use reflective mulch • Encourage ladybirds, hoverflies and wasps
TREATMENT AND CONTROL	• Spray with strong water jets, soap solution or neem • Use approved insecticides (e.g. pyrethrins, acetamiprid) if heavy, rotating products

5. Cutworms, Bagrada Bug and Seedling Pests

Insect pests

CAUSE AND SPREAD	Cutworms hide in soil by day and cut seedlings at night; Bagrada bugs (black shield bugs with orange marks) suck sap from young plants.
SIGNS AND SYMPTOMS	• Seedlings cut off at ground level; wilted young plants • Bagrada: white-pitted spots and wilting of seedlings, shrivelled growing points
PREVENTION	• Clear weeds before planting; deep ploughing exposes cutworms • Hand-pick cutworms; use collars around transplants • Plant on time; protect seedlings under net
TREATMENT AND CONTROL	• Apply approved insecticide or bait at the base; neem; spray Bagrada early in the morning • Replant gaps promptly

6. Downy Mildew and Alternaria Leaf Spot

Fungal / oomycete

CAUSE AND SPREAD	Cool, humid weather and leaf wetness promote infection; spores spread by wind and rain.
SIGNS AND SYMPTOMS	• Downy mildew: yellow patches on upper leaf surface, grey-white downy growth beneath; small black spots on head • Alternaria: dark, concentric ringed spots with yellow edges on older leaves
PREVENTION	• Use resistant varieties and clean seed; rotate; space plants; avoid overhead irrigation • Remove crop debris; improve air circulation
TREATMENT AND CONTROL	• Copper, mancozeb or chlorothalonil sprays; systemic fungicides in rotation per label • Remove severely infected leaves

7. Bacterial Soft Rot

Bacterial (Pectobacterium)

CAUSE AND SPREAD	Bacteria enter through wounds from insects, hail, cuts or other diseases; thrive in warm, wet conditions in the field and in storage.
SIGNS AND SYMPTOMS	• Slimy, wet, foul-smelling decay of the head or stalk; leaves collapse
PREVENTION	• Control caterpillars and other pests; avoid wounding plants • Ensure good drainage; harvest in dry weather; handle gently; store in cool, ventilated place • Rotate crops
TREATMENT AND CONTROL	• No cure; remove and discard rotten heads; disinfect tools and containers

8. Damping-off

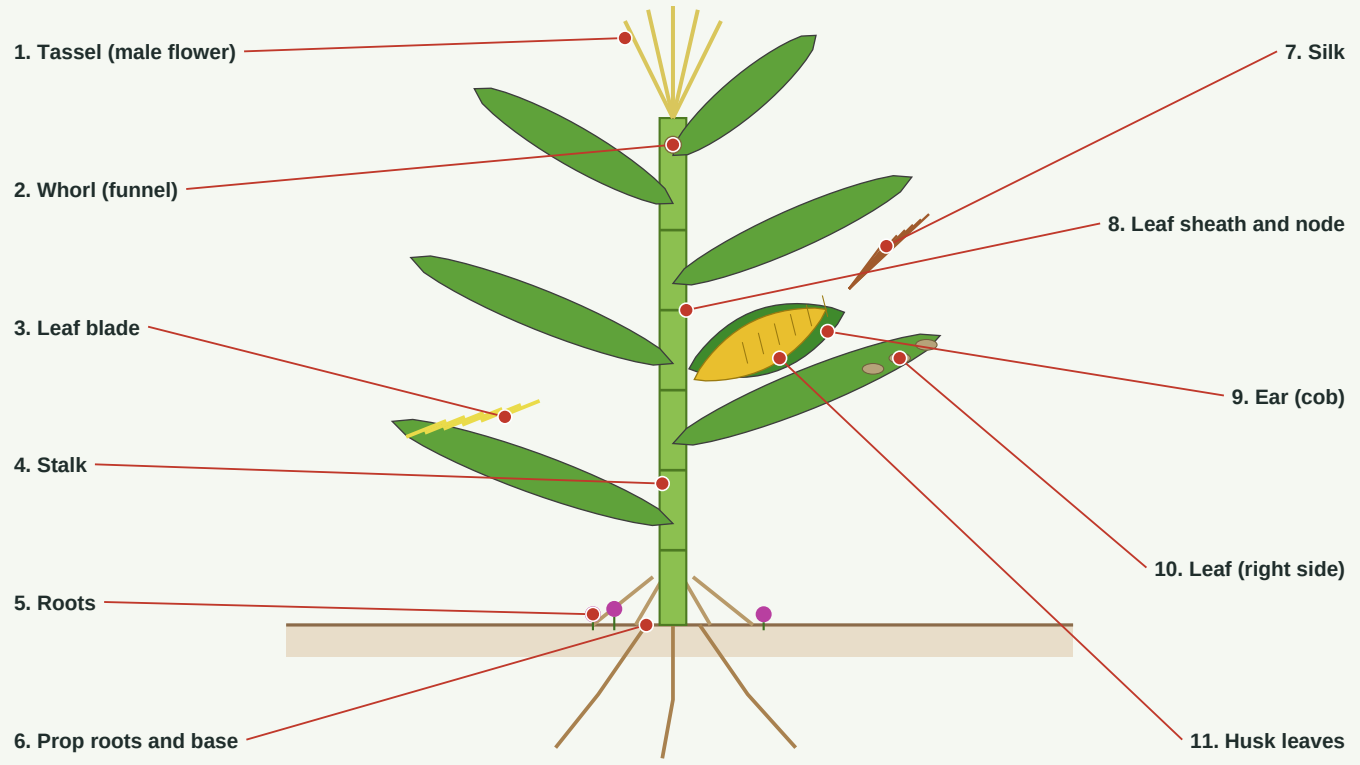
Fungal (Pythium, Rhizoctonia) - nursery disease

CAUSE AND SPREAD	Soil fungi attack seeds and seedlings in cold, wet, overcrowded nurseries.
SIGNS AND SYMPTOMS	• Seeds rot or seedlings fail to emerge • Seedlings collapse at soil level; stem pinched and discoloured ('wirestem')
PREVENTION	• Use clean seed and sterilised or solarised nursery soil; raised beds; avoid overwatering and overcrowding • Sow thinly; provide air movement
TREATMENT AND CONTROL	• Remove affected seedlings; improve drainage; apply fungicide drench per label (e.g. metalaxyl for Pythium)

B3. Maize

The main food crop of much of Africa; fall armyworm, stem borers, streak virus, striga and aflatoxin cause the greatest losses.

Figure 7. Maize plant - where to look for disease



Walk the field weekly from seedling stage: check the whorl, leaf blades, stalk, ear and roots.

No.	Body part / plant part	What to look for (and likely causes)
1	**Tassel (male flower)**	Armyworm feeding; white or reddish mould (smut); poor pollen shedding
2	**Whorl (funnel)**	Fresh sawdust-like frass and caterpillars (fall armyworm); dead heart (stem borer)
3	**Leaf blade**	Pale streaks (MSV); grey rectangular spots; cigar-shaped blight lesions; rust pustules
4	**Stalk**	Pin holes, tunnels, breakage (stem borers); soft stalk rot; lodging
5	**Roots**	Striga flowers (purple) near the base; poor, brown or eaten roots
6	**Prop roots and base**	Rot and lodging; ropy brown roots; striga attachment signs
7	**Silk**	Silks eaten or brown early (armyworm, drought); mould entry point
8	**Leaf sheath and node**	Stem borer entry holes; frass at nodes
9	**Ear (cob)**	Pink or green mould, rotted kernels, aflatoxin risk; black smut galls; insect holes
10	**Leaf (right side)**	Spots and lesions of leaf blight or grey leaf spot
11	**Husk leaves**	Torn or open husks letting in rain, birds and insects

1. Fall Armyworm (Spodoptera frugiperda)

Insect pest - caterpillars - major invasive pest across Africa

CAUSE AND SPREAD	Moths lay egg masses on leaves; larvae feed inside the whorl at night and spread fast; several generations per season.
SIGNS AND SYMPTOMS	• Ragged leaves with 'windowing' and shot holes in the whorl • Fresh sawdust-like frass (droppings) in the funnel • Caterpillars with an inverted 'Y' on the head and four dark dots on the second last body segment • Chewed tassels, silks and cobs; severe loss of young plants
PREVENTION	• Plant early and at the same time as neighbours; scout weekly from seedling stage • Intercrop with legumes; use push-pull (Desmodium and Napier grass); keep soil fertile • Rotate crops; deep ploughing exposes pupae; encourage natural enemies • Hand-crush egg masses and small larvae
TREATMENT AND CONTROL	• Early: put wood ash, fine sand, or soap-water into the whorl; neem extract or Bt • Approved insecticides (e.g. emamectin benzoate, spinetoram, chlorantraniliprole, lambda-cyhalothrin) applied into the whorl in the early morning or evening; rotate groups • Follow label, wear protective clothing, observe pre-harvest interval

2. Maize Streak Virus (MSV)

Viral - spread by leafhoppers (Cicadulina)

CAUSE AND SPREAD	Leafhoppers carry the virus from wild grasses and old maize to new crops; worst in late-planted and early-infected maize.
SIGNS AND SYMPTOMS	• Fine, broken, pale yellow streaks running along the leaf veins • Stunted plants; poor or no cobs; yield loss up to 100 percent if infected early
PREVENTION	• Plant early; use tolerant or resistant varieties; control grass weeds • Remove and destroy infected plants early; avoid planting next to old maize • Seed dressing with an insecticide can protect seedlings
TREATMENT AND CONTROL	• No cure; rogue infected plants • Control leafhoppers where numbers are high

3. Maize Lethal Necrosis (MLN)

Viral (MCMV + SCMV) - emerging in East and Central Africa

CAUSE AND SPREAD	Caused by two viruses together, spread by thrips, beetles, aphids and infected seed or soil debris.
SIGNS AND SYMPTOMS	• Yellow chlorotic mottle, leaf-edge drying ('dead heart') • Premature plant death; small malformed cobs with little grain
PREVENTION	• Plant certified, treated seed and tolerant varieties; rotate with non-cereal crops • Control insect vectors and weeds; leave maize-free periods • Roguing and destroying infected plants early
TREATMENT AND CONTROL	• No cure; uproot and destroy infected plants • Report to extension officers

4. Grey Leaf Spot

Fungal (Cercospora zeae-maydis)

CAUSE AND SPREAD	Fungus survives in maize debris; favoured by warm, humid weather and minimum tillage.
SIGNS AND SYMPTOMS	• Narrow, rectangular grey or tan lesions between leaf veins • Lesions merge, causing leaf death; reduced grain filling and weak stalks
PREVENTION	• Rotate maize with non-cereal crops; bury residue; use resistant varieties; adequate spacing • Plant early
TREATMENT AND CONTROL	• Foliar fungicides (strobilurin or triazole mixtures) at tasselling if severe, per label

5. Northern Leaf Blight and Common Rust

Fungal (Exserohilum turcicum; Puccinia sorghi)

CAUSE AND SPREAD	Wind-borne spores in cool, humid weather.
SIGNS AND SYMPTOMS	• Blight: long, cigar-shaped grey-green to tan lesions 2.5-15 cm long • Rust: small cinnamon or brick-red powdery pustules on both leaf surfaces
PREVENTION	• Resistant varieties; rotate crops; remove debris; early planting
TREATMENT AND CONTROL	• Fungicide sprays (triazole or strobilurin) if early and severe, per label

6. Maize Stem Borers (Busseola fusca, Chilo partellus, Sesamia)

Insect pests - caterpillars

CAUSE AND SPREAD	Moths lay eggs on leaves; larvae bore into stems and cobs. Can destroy up to 50 percent of the crop.
SIGNS AND SYMPTOMS	• Small pinholes and 'windowing' on young leaves • Dead hearts (central leaf dies and pulls out easily) • Tunnels, frass and broken stalks; damaged cobs
PREVENTION	• Push-pull intercrop; destroy stalks after harvest; early planting; crop rotation • Use Bt or resistant varieties where allowed; encourage natural enemies (e.g. Cotesia wasps)
TREATMENT AND CONTROL	• Apply approved insecticide granules or spray into the whorl at 3-4 weeks (e.g. cypermethrin, lambda-cyhalothrin) • Remove and burn infested stalks

7. Ear Rots and Aflatoxin

Fungal (Aspergillus flavus, Fusarium) - food safety danger

CAUSE AND SPREAD	Fungi enter the cob through damaged husks, insect holes and bird damage; worsened by drought stress and wet harvest, and by poor drying and storage. Aspergillus produces aflatoxin, which causes liver damage, cancer and stunting in children.
SIGNS AND SYMPTOMS	• Greenish-yellow powdery mould (Aspergillus); pink or white mould (Fusarium) • Mouldy, discoloured kernels at the cob tip; musty smell • Contaminated grain may look normal - aflatoxin cannot be seen
PREVENTION	• Plant on time with good fertility; control stem borers and birds; harvest at maturity • Dry grain quickly on tarpaulins to 13 percent moisture; sort and remove mouldy cobs • Store off the ground in clean, dry stores or hermetic bags; use biocontrol (Aflasafe) where available
TREATMENT AND CONTROL	• No cure; discard mouldy grain - do not eat or feed to animals • Sort and remove contaminated kernels before storage

8. Striga (Witchweed)

Parasitic weed (Striga hermonthica, S. asiatica) - drylands

CAUSE AND SPREAD	Tiny seeds persist in soil for many years and germinate near maize roots; the parasite attaches to roots and steals water and nutrients before it even emerges.
SIGNS AND SYMPTOMS	• Stunted, yellowing, scorched, wilting maize despite adequate rain • Purple, pink or white striga flowers at the base of maize plants • Poor cob formation
PREVENTION	• Rotate with trap crops (soybean, cowpea, groundnut); intercrop with Desmodium • Apply manure and fertiliser; use striga-tolerant or imazapyr-resistant maize varieties • Hand-pull striga before it flowers; use clean seed; clean tools
TREATMENT AND CONTROL	• Hand-weed before seed set and burn the weeds • Use imazapyr-coated seed (IR maize) where recommended

9. Stored Grain Pests (Larger Grain Borer, Maize Weevil)

Insect pests - storage

CAUSE AND SPREAD	Beetles carried from field or store; the larger grain borer (Prostephanus truncatus) is very destructive.
SIGNS AND SYMPTOMS	• Holes in grain; powdery flour at the bottom; warm, musty smell • Live beetles; grain loses weight and quality
PREVENTION	• Harvest dry; sun-dry grain; clean stores and bags; use hermetic storage bags • Do not mix old and new grain; inspect regularly
TREATMENT AND CONTROL	• Sun-dry again and sieve; admix with approved insecticide or ash/neem; trained operators may fumigate stores

10. Common Smut (Ustilago maydis)

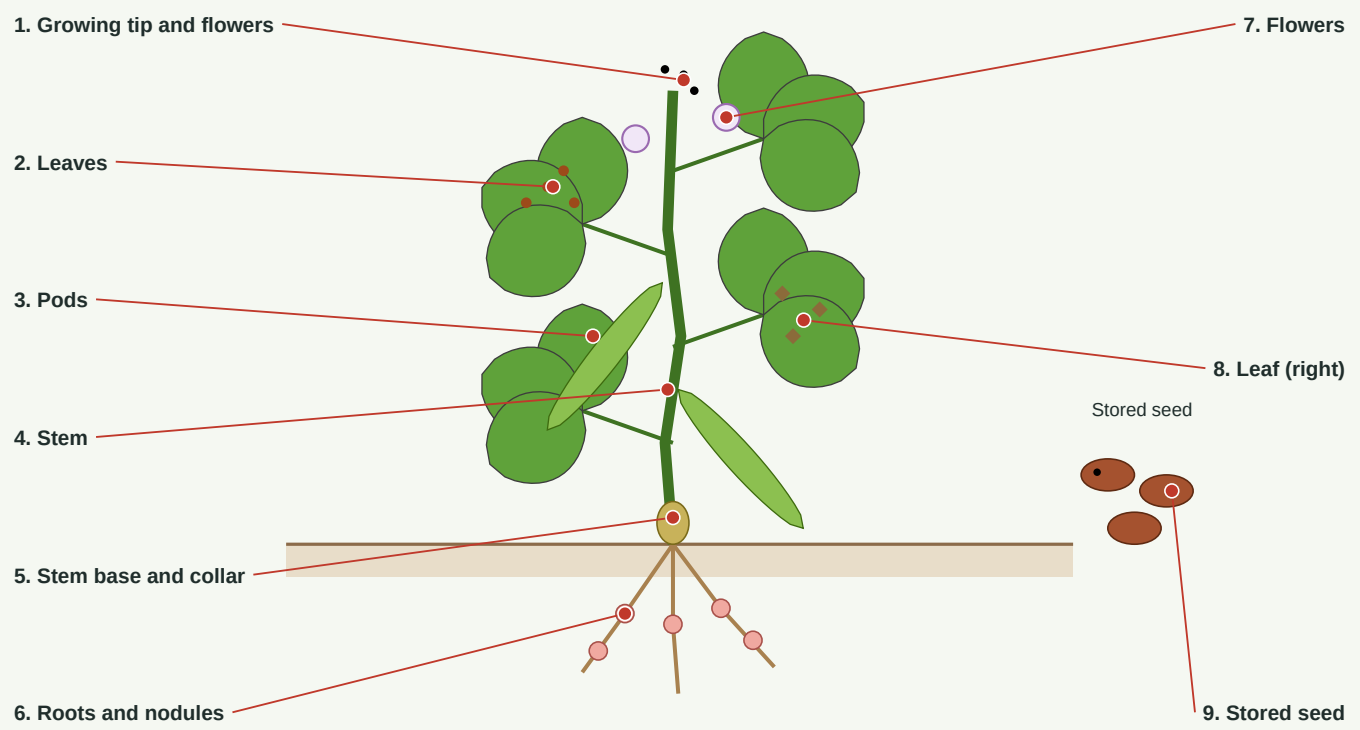
Fungal

CAUSE AND SPREAD	Spores in soil and debris enter through wounds.
SIGNS AND SYMPTOMS	• Silvery-white galls on cobs, tassel or stem that burst into black powder
PREVENTION	• Rotate crops; avoid injury to plants; use resistant varieties
TREATMENT AND CONTROL	• Remove and bury galls before they burst; no chemical control needed

B4. Common Beans

Beans improve soil nitrogen but are prone to leaf spots, root rot and storage pests.

Figure 8. Bean plant - where to look for disease



Check leaves (both sides), stems near the soil, pods and roots for pink healthy nodules.

No.	Body part / plant part	What to look for (and likely causes)
1	**Growing tip and flowers**	Black or green aphids; curled, mosaic or distorted leaves; flower drop
2	**Leaves**	Angular brown spots between veins; yellow mosaic; halo blight spots; rust pustules
3	**Pods**	Sunken dark spots (anthracnose); water-soaked spots; holes and frass (pod borers)
4	**Stem**	Dark, sunken cankers (anthracnose); tunnels and maggots near the base
5	**Stem base and collar**	Swollen, cracked base with maggots; red-brown root rot lesions
6	**Roots and nodules**	Pink nodules show healthy nitrogen fixing; brown rotted roots; nematode galls
7	**Flowers**	Flower drop from heat or thrips damage
8	**Leaf (right)**	Small reddish rust pustules on underside; leaf spots
9	**Stored seed**	Round exit holes and powdery dust (bean bruchids)

1. Angular Leaf Spot

Fungal (Pseudocercospora griseola)

CAUSE AND SPREAD	Seed-borne and survives in debris; spreads by rain splash and wind in warm, humid weather.
SIGNS AND SYMPTOMS	• Angular grey-brown spots bounded by leaf veins; yellowing and early leaf drop • Dark spots on pods; discoloured seeds
PREVENTION	• Use clean certified seed and resistant varieties; rotate with cereals for 2 years • Avoid working in wet fields; bury residue; intercrop with maize
TREATMENT AND CONTROL	• Copper, mancozeb or chlorothalonil sprays at first signs per label; remove infected plants

2. Anthracnose

Fungal (Colletotrichum lindemuthianum)

CAUSE AND SPREAD	Seed-borne; spreads by rain splash and by people and animals moving through wet crops.
SIGNS AND SYMPTOMS	• Dark brick-red to black sunken lesions along veins on leaf undersides • Sunken, dark-edged cankers on stems and pods; pink spore masses in wet weather • Discoloured seed; seedling death
PREVENTION	• Use clean seed and resistant varieties; rotate with non-legumes; bury debris • Do not work in wet fields; wide spacing
TREATMENT AND CONTROL	• Mancozeb, copper or azoxystrobin sprays per label; remove infected plants

3. Common Bacterial Blight and Halo Blight

Bacterial (Xanthomonas, Pseudomonas) - seed-borne

CAUSE AND SPREAD	Spread by seed, rain splash, wind-blown rain and tools.
SIGNS AND SYMPTOMS	• Water-soaked spots that turn brown with a yellow border (halo) • Leaf edges scorch; pods spotted and wrinkled; wilting
PREVENTION	• Use certified seed; rotate; avoid working in wet fields; remove debris • Plant resistant or tolerant varieties
TREATMENT AND CONTROL	• Copper-based bactericides can slow spread; rogue badly infected plants

4. Bean Rust

Fungal (Uromyces appendiculatus)

CAUSE AND SPREAD	Wind-borne spores; favoured by cool nights and humid days.
SIGNS AND SYMPTOMS	• Small, rust-coloured powdery pustules mainly on the underside of leaves • Yellowing, leaf drop; reduced pod filling
PREVENTION	• Resistant varieties; early planting; rotation; wide spacing; remove debris
TREATMENT AND CONTROL	• Sulphur, mancozeb or tebuconazole sprays per label

5. Bean Common Mosaic Virus (BCMV)

Viral - aphid-borne and seed-borne

CAUSE AND SPREAD	Spread by aphids and infected seed; some strains cause black root.
SIGNS AND SYMPTOMS	• Light and dark green mosaic mottling; leaf curling and puckering • Stunted plants; poor pods; dead tip (black root strain)
PREVENTION	• Use virus-free certified seed; plant resistant varieties; control aphids and weeds • Uproot infected plants early
TREATMENT AND CONTROL	• No cure; rogue infected plants; control aphids

6. Root Rots (Fusarium, Pythium, Rhizoctonia)

Soil-borne fungi

CAUSE AND SPREAD	Soil fungi attack roots of stressed plants in waterlogged, compacted or infertile soils.
SIGNS AND SYMPTOMS	• Yellowing, wilting, stunting • Reddish-brown or black rot of the tap root and base of stem; poor stand
PREVENTION	• Rotate with cereals; plant on raised beds or ridges; improve drainage and organic matter • Avoid compaction and waterlogging; use treated seed; apply phosphorus fertiliser
TREATMENT AND CONTROL	• No cure; earth-up stems to promote new roots; remove badly affected plants

7. Bean Stem Maggot (Ophiomyia)

Insect pest - flies whose larvae bore into the stem

CAUSE AND SPREAD	Small shiny black flies lay eggs in seedling leaves; larvae tunnel into the stem base within 1-3 weeks of emergence.
SIGNS AND SYMPTOMS	• Yellowing, wilting and death of young seedlings • Swollen, cracked, discoloured stem base with small tunnels and maggots • Seedlings easily uprooted
PREVENTION	• Plant early; mulch; improve soil fertility with manure; avoid overcrowding • Earth-up stems; use insecticide-treated seed (where recommended); rotate
TREATMENT AND CONTROL	• Earth up the base of the stem to promote adventitious roots • Spray lambda-cyhalothrin or dimethoate per label soon after emergence if severe; apply neem as an alternative

8. Aphids and Whiteflies

Insect pests - sap-sucking

CAUSE AND SPREAD	Black bean aphids and whiteflies colonise growing tips and leaf undersides and spread viruses.
SIGNS AND SYMPTOMS	• Clusters of small black or green insects on young shoots • Curled, yellow leaves; honeydew and sooty mould; stunted plants
PREVENTION	• Inspect regularly; control weeds; intercrop; encourage ladybirds • Plant resistant varieties
TREATMENT AND CONTROL	• Soap spray, neem, or pyrethrin-based insecticide; use approved insecticides in rotation per label

9. Bean Bruchids and Storage Weevils

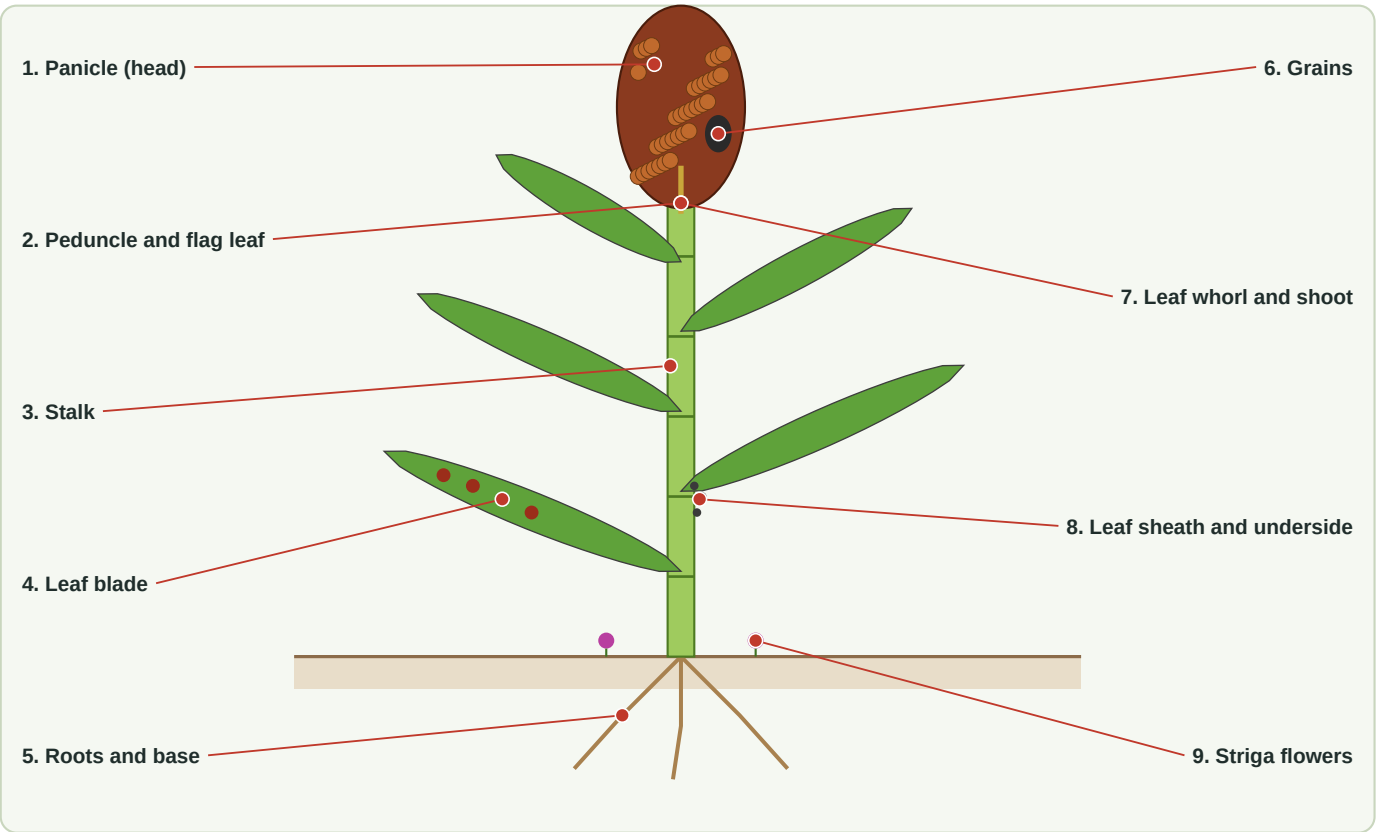
Insect pests - storage (Acanthoscelides, Zabrotes)

CAUSE AND SPREAD	Beetles lay eggs on pods in the field or on stored seed; larvae feed inside the seeds.
SIGNS AND SYMPTOMS	• Round exit holes in seeds; powder and dust in stored beans; live beetles • Light, hollow, wasted seed
PREVENTION	• Harvest dry beans; sun-dry; clean stores; use hermetic bags • Mix with ash, sand or neem leaves; do not mix old and new seed
TREATMENT AND CONTROL	• Solarise infested seed in the sun in clear plastic; freeze or heat seed; admix approved storage insecticide • Do not eat seed treated with insecticide intended for planting

B5. Sorghum

A hardy, drought-tolerant grain; threatened by grain mould, smuts, striga, shoot fly, stem borers, midge and birds.

Figure 9. Sorghum plant - where to look for disease



Inspect seedlings for dead hearts, leaves for spots, stalks for tunnels and heads for mould, smut or insects.

No.	Body part / plant part	What to look for (and likely causes)
1	**Panicle (head)**	Black smut sacs, mould, ergot honeydew, midge-chaffy florets, bird damage
2	**Peduncle and flag leaf**	Brown rot or breakage of the neck below the head (stalk rot, borers)
3	**Stalk**	Red or purple inner tissue (red rot); borer tunnels and holes; lodging
4	**Leaf blade**	Red or purple spots with dark margins (anthracnose); tan blight lesions; rust pustules
5	**Roots and base**	Striga attachment, root rot, poor roots
6	**Grains**	Mouldy, chaffy, light, eaten by birds or midge
7	**Leaf whorl and shoot**	Dead heart that pulls out easily (shoot fly, stem borer)
8	**Leaf sheath and underside**	Aphid colonies, honeydew, sooty mould, red leaves
9	**Striga flowers**	Purple or pink flowers around the base of the plant

1. Grain Mould

Fungal complex (Fusarium, Curvularia, Phoma)

CAUSE AND SPREAD	Fungi attack developing grain when rain falls during grain fill and ripening; compact heads and white grain are more susceptible.
SIGNS AND SYMPTOMS	• White, pink, grey or black mould on grains • Chaffy, light, discoloured grain; poor germination; lower food quality
PREVENTION	• Plant early-maturing or open-headed, tannin-rich or coloured grain varieties • Time planting so that grain filling falls in dry weather; harvest at maturity; dry quickly
TREATMENT AND CONTROL	• No effective chemical control; remove and discard mouldy heads; store only dry, clean grain

2. Smuts (Covered Kernel, Loose and Head Smut)

Fungal (Sporisorium sorghi, S. cruentum, S. reilianum)

CAUSE AND SPREAD	Spores on seed or in soil infect seedlings; fungus grows inside the plant until the head forms.
SIGNS AND SYMPTOMS	• Covered kernel smut: grains replaced by grey-white, hard spore sacs • Loose smut: sacs rupture into black powder • Head smut: entire head replaced by black mass with fibrous threads; plants stunted
PREVENTION	• Plant certified seed dressed with approved fungicide (e.g. thiram or carboxin); rotate crops • Remove and burn smutted heads before they burst; use resistant varieties
TREATMENT AND CONTROL	• Seed dressing is the main control; roguing smutted plants; avoid saving seed from infected fields

3. Ergot (Sugary Disease)

Fungal (Claviceps africana)

CAUSE AND SPREAD	Fungus infects flowers during humid, cool weather at flowering; spores spread by wind and rain.
SIGNS AND SYMPTOMS	• Sticky, sweet honeydew droplets from florets, often turning white or black (sooty mould) • Hard dark ergot bodies replace grains; poor seed set • Birds and insects attracted to honeydew
PREVENTION	• Plant early so flowering coincides with drier conditions; avoid late flowering and staggered planting • Plant resistant varieties; remove infected heads
TREATMENT AND CONTROL	• Control is mainly cultural; discard heavily affected heads and do not feed them to animals

4. Anthracnose

Fungal (Colletotrichum sublineolum)

CAUSE AND SPREAD	Spores survive on residue and seed; spread by rain splash in warm, humid weather.
SIGNS AND SYMPTOMS	• Circular red, purple or tan spots with dark margins on leaves; tiny black dots in the centres • Midrib red rot; stalk rot; reddish discolouration of stalk; head and grain infection
PREVENTION	• Use clean seed and resistant varieties; rotate with legumes; bury residue; wide spacing
TREATMENT AND CONTROL	• Foliar fungicides (azoxystrobin, mancozeb) if severe, per label; remove infected residue

5. Leaf Blight, Rust and Leaf Spots

Fungal (Exserohilum, Puccinia purpurea, Cercospora)

CAUSE AND SPREAD	Wind-borne spores in humid conditions.
SIGNS AND SYMPTOMS	• Blight: long, tan, cigar-shaped lesions on leaves • Rust: reddish-purple pustules on both leaf surfaces • Leaf spots: grey to purple spots with yellow borders
PREVENTION	• Resistant varieties; rotate; remove debris; early planting; balanced fertilisation
TREATMENT AND CONTROL	• Foliar fungicide only if severe and economic, per label

6. Striga (Witchweed)

Parasitic weed (Striga hermonthica) - drought-prone areas

CAUSE AND SPREAD	Striga seeds germinate near sorghum roots and attach to steal water and nutrients.
SIGNS AND SYMPTOMS	• Stunted, yellow, scorched, wilting sorghum; reduced head size • Purple or pink striga flowers around the base of plants
PREVENTION	• Rotate with soybean, cowpea or groundnut; apply manure or fertiliser; use tolerant varieties • Hand-pull striga before flowering; use clean seed; intercrop with Desmodium
TREATMENT AND CONTROL	• Hand-weed before seed set and burn; resistant or tolerant varieties and improved soil fertility

7. Sorghum Shoot Fly (Atherigona soccata)

Insect pest - seedling pest

CAUSE AND SPREAD	Small flies lay eggs on seedling leaves; larvae bore into the shoot and kill the growing point.
SIGNS AND SYMPTOMS	• Central leaf dries to form a 'dead heart' that pulls out easily and smells bad • Tillers sprout and develop poorly; reduced stand
PREVENTION	• Plant early at the onset of rain; use a higher seed rate and thin later; apply fertiliser • Seed dressing with insecticide where recommended; use resistant varieties
TREATMENT AND CONTROL	• Remove dead hearts and bury; spray cypermethrin or dimethoate on seedlings if severe per label

8. Sorghum Stem Borers

Insect pests (Chilo partellus, Busseola fusca)

CAUSE AND SPREAD	Moths lay eggs on leaves; larvae feed in the whorl then bore into the stem.
SIGNS AND SYMPTOMS	• Windowing on young leaves; dead heart • Tunnels in stems, frass; broken stalks, chaffy or lost panicles
PREVENTION	• Early planting; destroy stalks; rotation; push-pull intercropping; natural enemies • Plant resistant varieties
TREATMENT AND CONTROL	• Apply approved insecticide granules or sprays into the whorl; remove infested stalks

9. Sorghum Midge (Stenodiplosis sorghicola)

Insect pest - flowering stage

CAUSE AND SPREAD	Tiny orange flies lay eggs in flowers; larvae feed on developing grain.
SIGNS AND SYMPTOMS	• Empty, chaffy spikelets with small white pupal cases • Poor seed set; head looks 'blind' despite normal flowering
PREVENTION	• Plant uniformly and early so that crop flowers together; avoid staggered planting • Remove wild sorghum hosts; use resistant varieties
TREATMENT AND CONTROL	• Spray at flowering in the early morning with approved pyrethroid if midge counts are high

10. Sugarcane Aphid (Melanaphis sacchari)

Insect pest - sap-sucking

CAUSE AND SPREAD	Colonies on leaf undersides produce honeydew; build up in dry, warm conditions.
SIGNS AND SYMPTOMS	• Sticky honeydew and black sooty mould on leaves and heads • Leaves yellow, redden and dry; stunted plants; poor heads
PREVENTION	• Scout regularly; control weeds; protect natural enemies; plant resistant or tolerant varieties
TREATMENT AND CONTROL	• Soap or neem spray; use approved insecticides on local advice if infestation is heavy

11. Quelea Birds and Other Bird Damage

Vertebrate pest (Quelea quelea)

CAUSE AND SPREAD	Large flocks feed on ripening grain and can destroy a crop in a few days; worst on late-maturing and small fields.
SIGNS AND SYMPTOMS	• Partly eaten or empty heads; broken stalks and scattered grain
PREVENTION	• Plant together with neighbours at the same time; use bird-resistant (tannin) or compact head varieties • Bird scaring by children or noise; harvest promptly; community-level bird control by authorities
TREATMENT AND CONTROL	• Rely on scaring and early harvest; organised control is done by agricultural authorities only

Quick Symptom Lookup Tables

Use these as a starting point; always confirm with a vet or extension officer.

Livestock: sign - possible diseases

Sign	Chickens / Goats	Pigs / Cattle
Sudden death	Newcastle, avian influenza, fowl cholera; enterotoxaemia, heartwater	ASF, anthrax, blackleg, RVF, redwater
Bloody or watery diarrhoea	Coccidiosis, Gumboro; PPR, coccidiosis, worms	ASF, CSF, E. coli; FMD, trypanosomiasis, worms
Cough, laboured breathing	Newcastle, bronchitis; CCPP, PPR, pneumonia	Pneumonia, ASF; CBPP, TB, pneumonia
Mouth blisters, ulcers, drooling	Fowl pox (mouth); PPR, orf	FMD; FMD, mouth lesions
Skin lumps or nodules	Fowl pox; orf, caseous lymphadenitis	Erysipelas (diamond lesions); LSD, dermatophilosis
Pale eyelids, anaemia	Mites, worms; haemonchosis, ticks	Piglet anaemia, worms; ECF, babesiosis, trypanosomiasis
Abortion	-; brucellosis, CCPP	ASF, CSF, brucellosis; brucellosis, RVF
Nervous signs (circling, twitching)	Newcastle, Marek's; heartwater, listeriosis	CSF; heartwater, cerebral babesiosis, rabies
Lameness	Marek's; foot rot, arthritis	FMD, erysipelas; FMD, blackleg, foot rot

Crops: sign - possible causes

Sign	Possible causes	First action
Sudden wilting, green leaves	Bacterial wilt, Fusarium wilt, nematodes, stem borers, root rot	Cut stem for brown tissue; check roots; uproot and destroy; rotate
Leaf spots with rings or blotches	Early blight, late blight, grey leaf spot, anthracnose, angular leaf spot	Remove infected leaves; fungicide; improve spacing
Curled or yellow mottled leaves	TYLCV, maize streak virus, bean mosaic, MLN, aphids	Control insect vectors; rogue plants; use resistant seed
Holes in leaves, frass	Fall armyworm, diamondback moth, stem borers, bollworm, bagrada	Scout; Bt or approved insecticide; hand-pick
Swollen or galled roots	Root-knot nematodes, clubroot	Rotate; lime acid soil; destroy roots
Fruit, cob or grain rot or mould	Blossom end rot, late blight, ear rots (aflatoxin), grain mould	Dry grain; sort mouldy produce; adjust watering
Stunting with parasitic flowers at the base	Striga on maize and sorghum	Hand-weed before seeding; rotate with legumes
Holes and dust in stored grain	Larger grain borer, maize weevil, bean bruchids	Dry grain, use hermetic bags, clean stores

Final reminder

- Disease patterns, available products, doses and laws differ between countries. Always check with your national Veterinary Services, Ministry of Agriculture, an accredited vet or extension officer.
- Report unusual deaths and notifiable diseases early - early reporting protects your farm, your neighbours and the nation.